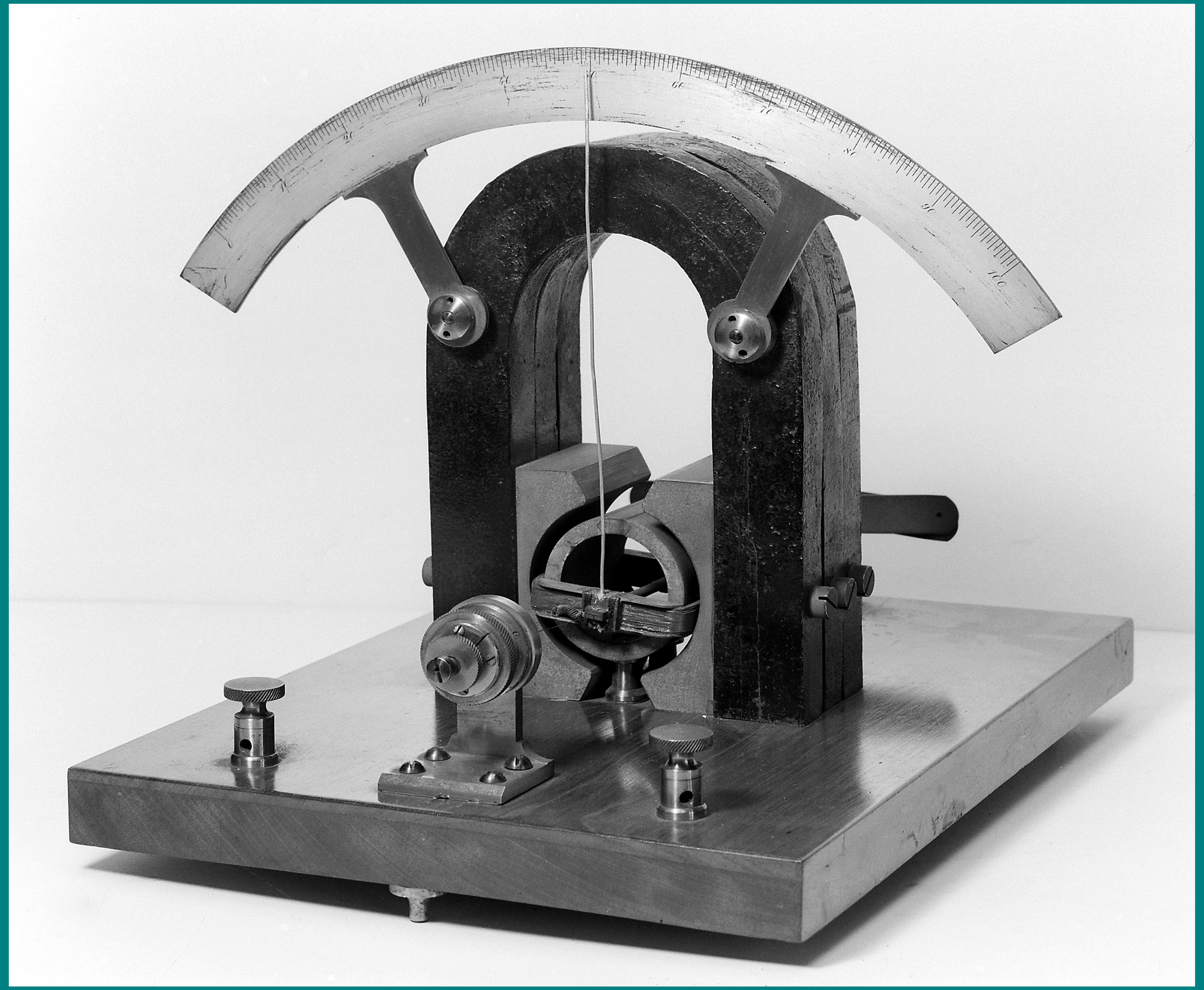
THE MOVING COIL GALVANOMETER



Moving Coil Galvanometer (MCG)

- A **moving coil galvanometer** is an **instrument** which is **used to detect and measure electric currents**.

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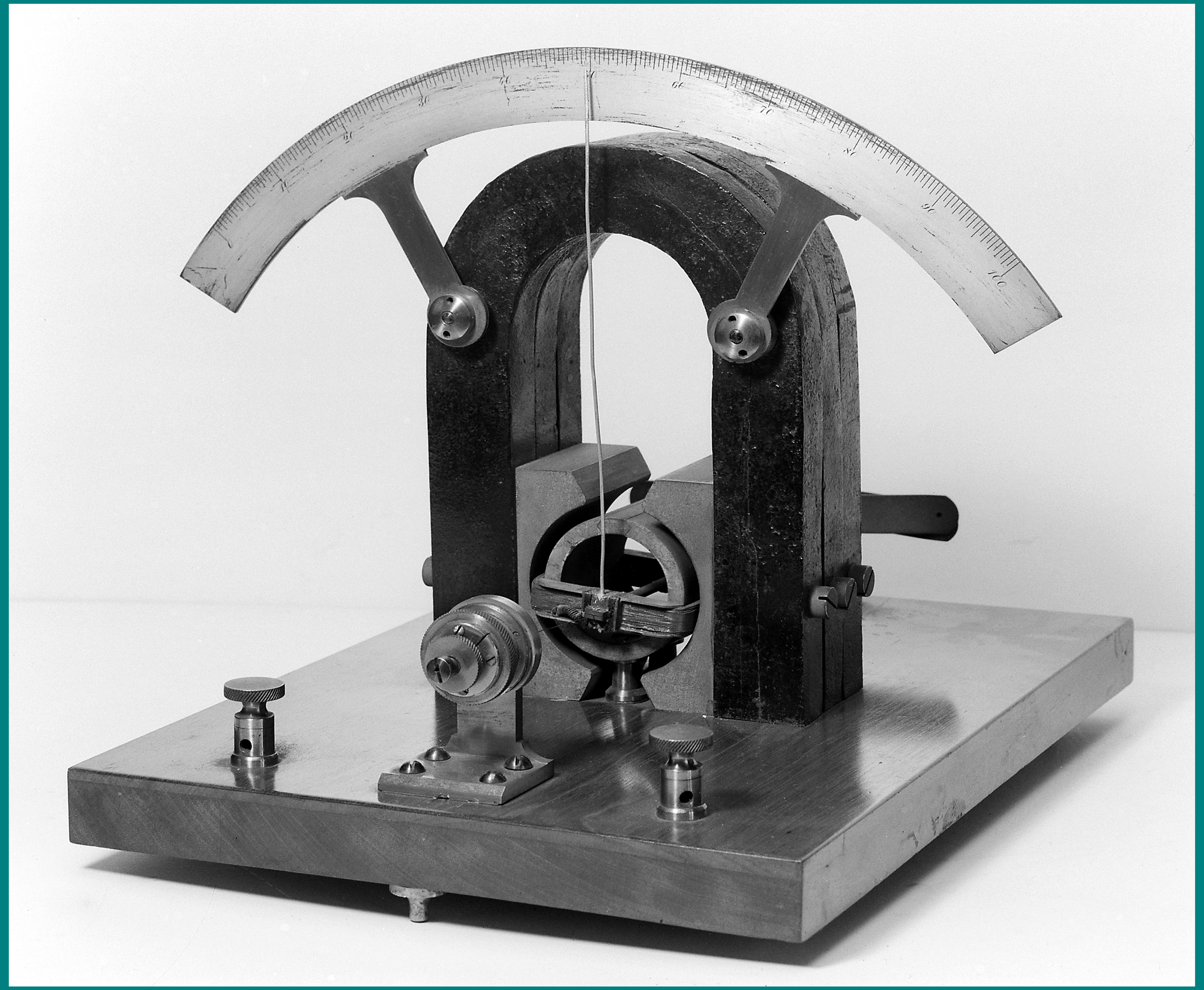

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Working Principle of A MCG

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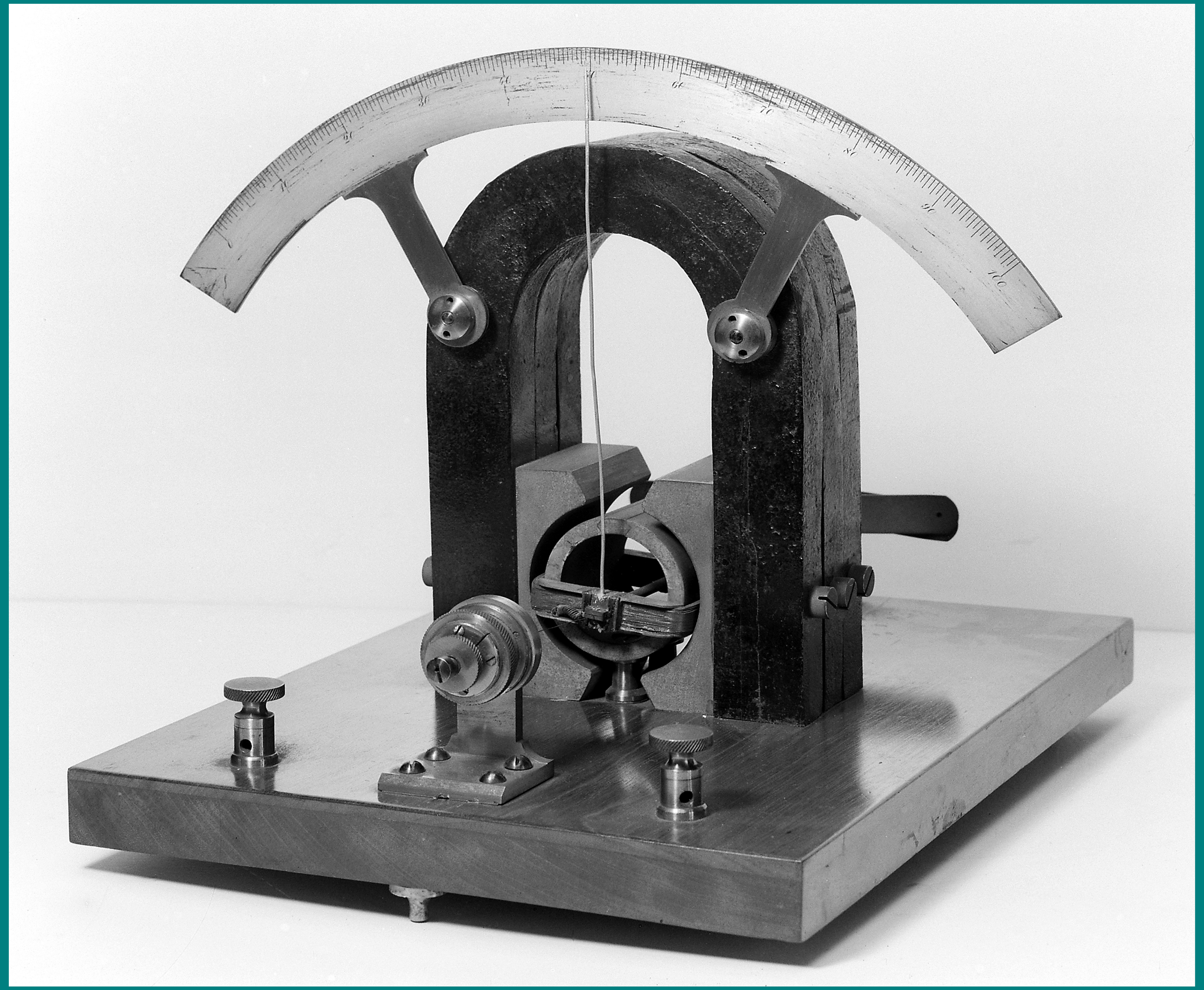

Moving Coil Galvanometer (MCG)

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Working Principle of A MCG

- A **current-carrying coil**, when **placed in** an **external magnetic field**, **experiences** a **torque**, **which deflects** the **coil**. and the **deflection is directly proportional to the amount of the current** passed through the coil.

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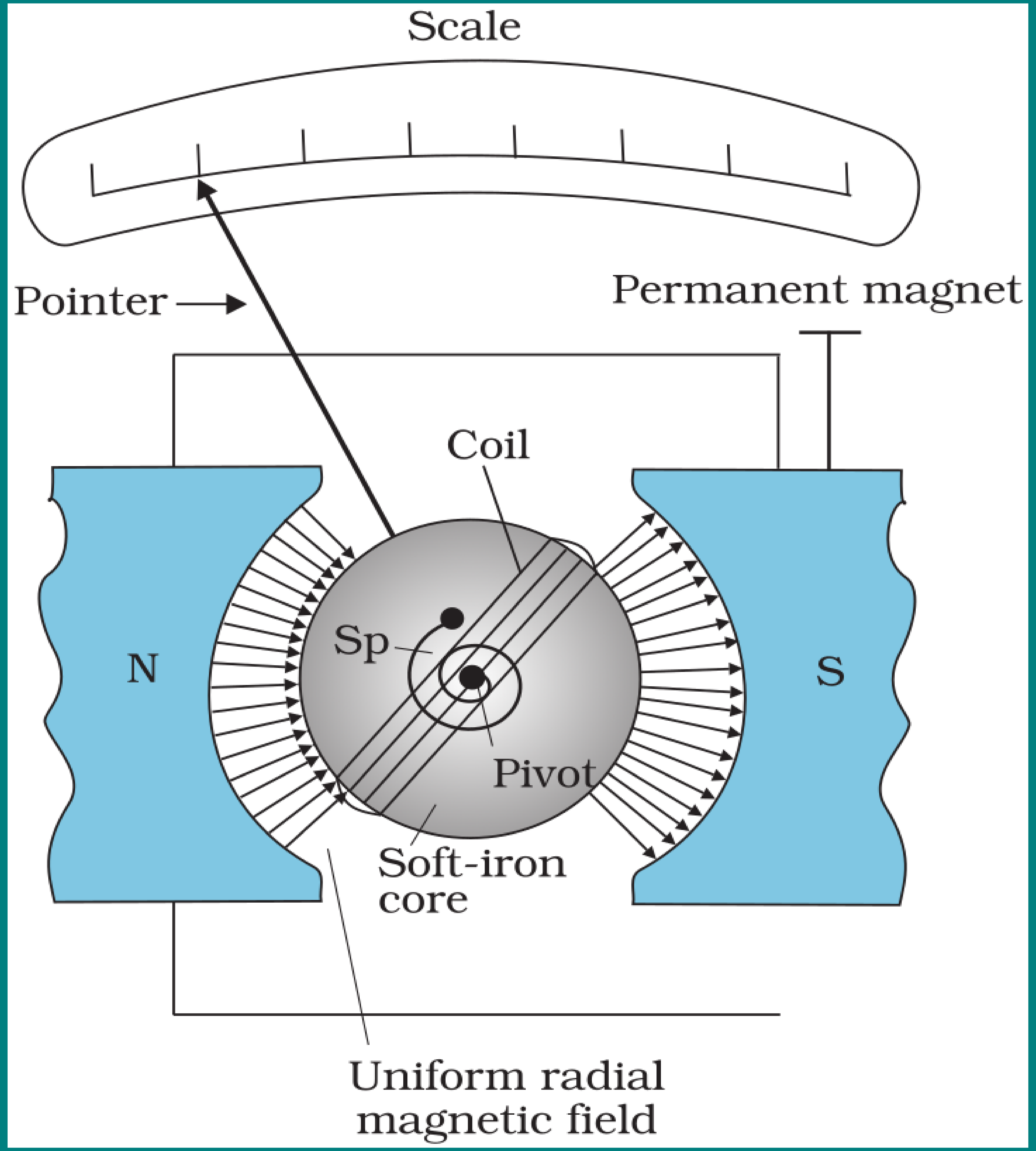

Construction of A (MCG)

- The **galvanometer consists of a coil**, with **many turns (N)**, **free to rotate** about a fixed axis in a **uniform radial magnetic field** B

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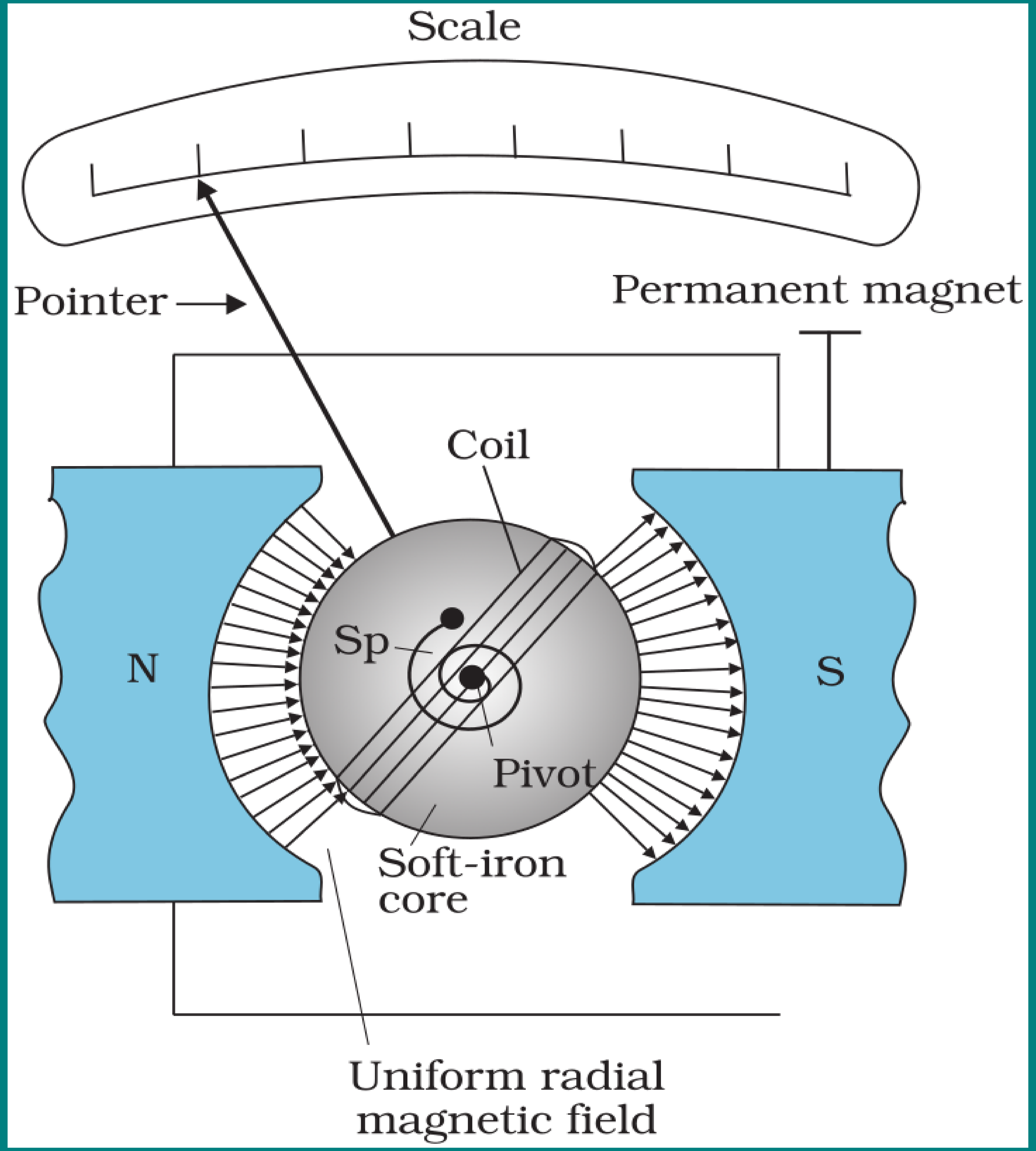

Construction of A (MCG)

- The **galvanometer consists of a coil**, with **many turns (N)**, **free to rotate** about a fixed axis in a **uniform radial magnetic field** B
- There is a **cylindrical soft iron core** which not only **makes the field radial** but **also increases the strength of the magnetic field**.

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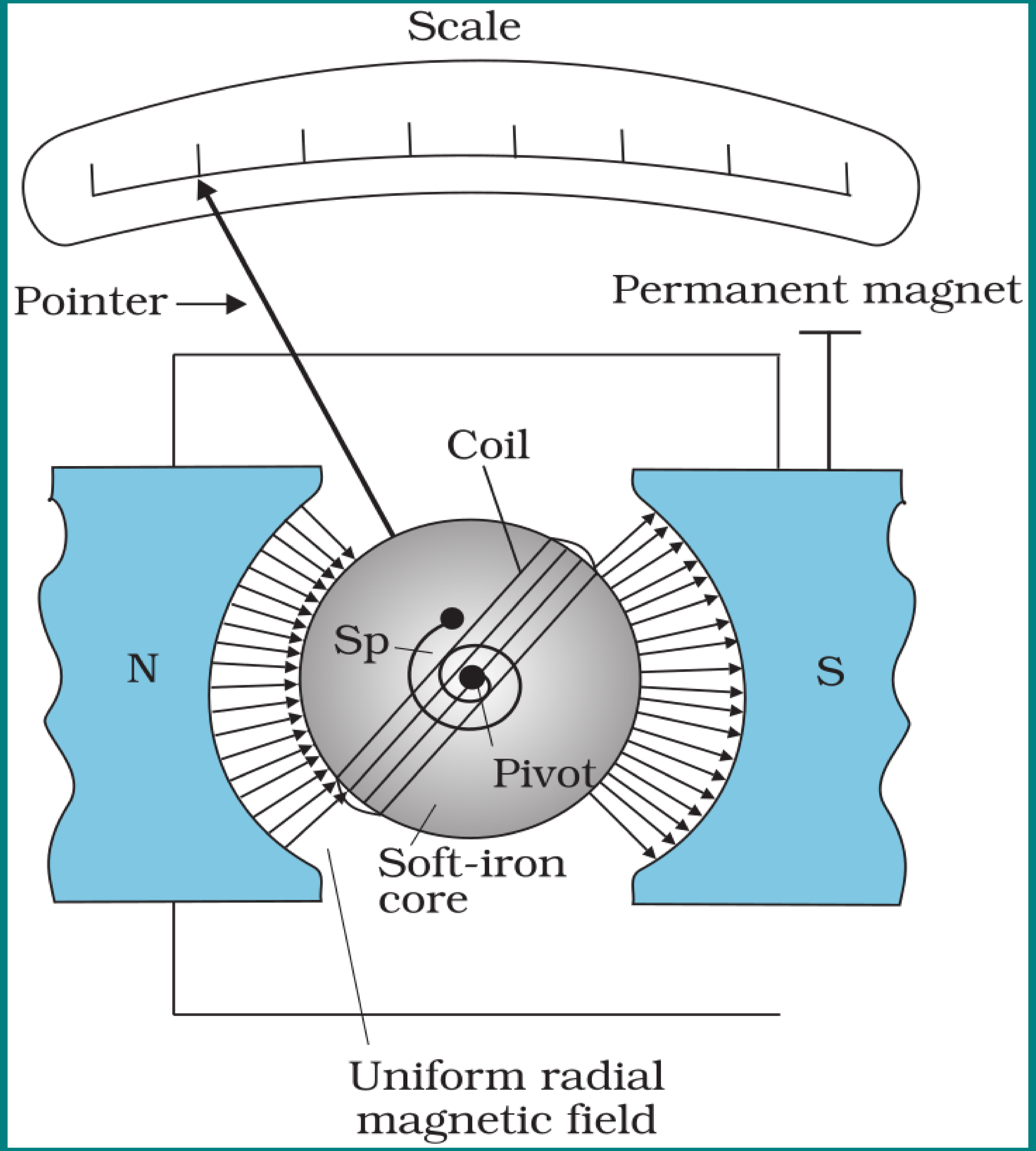

Construction of A (MCG)

- The **galvanometer consists of a coil**, with **many turns (N)**, **free to rotate** about a fixed axis in a **uniform radial magnetic field** B
- There is a **cylindrical soft iron core** which not only **makes the field radial** but **also increases the strength of the magnetic field**.
- The **lower end of the coil is connected to a spring** (S_P)

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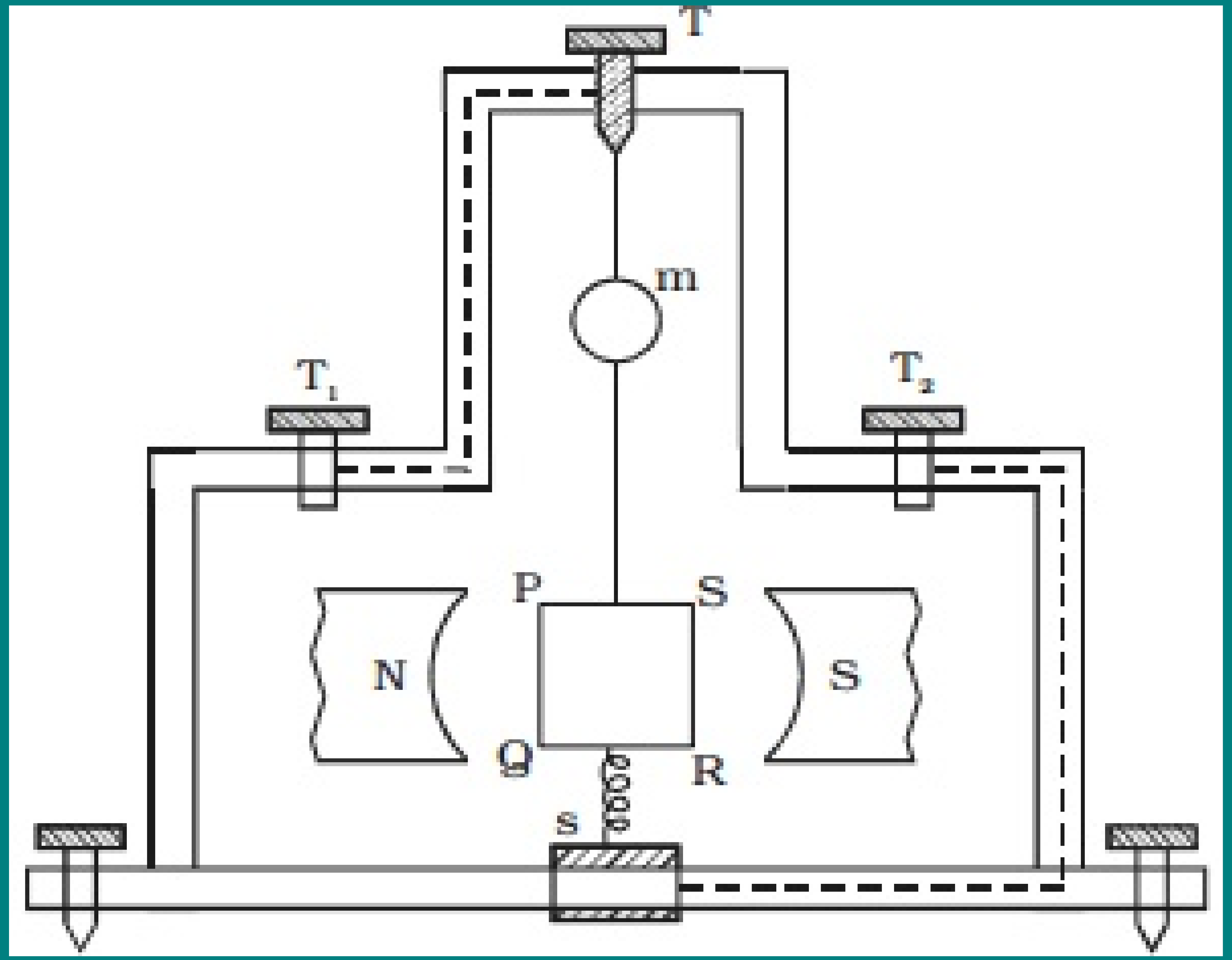

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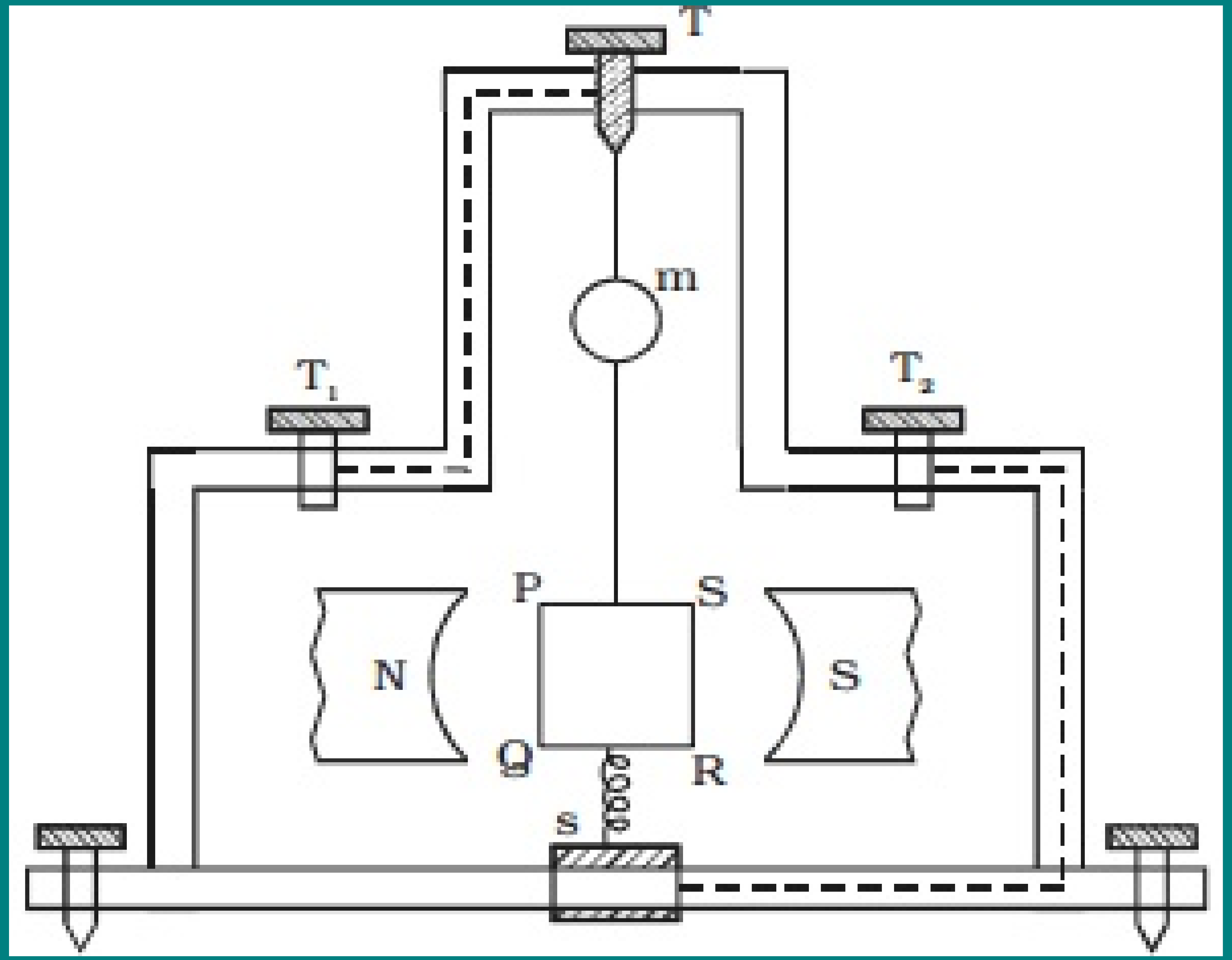

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- **When** a **current (I) flows** through the coil, a **torque acts on it**. This torque is given by $\boldsymbol{\tau} = \boldsymbol{NIAB}$ ($\sin\theta = 1$ since the filed is radial.)

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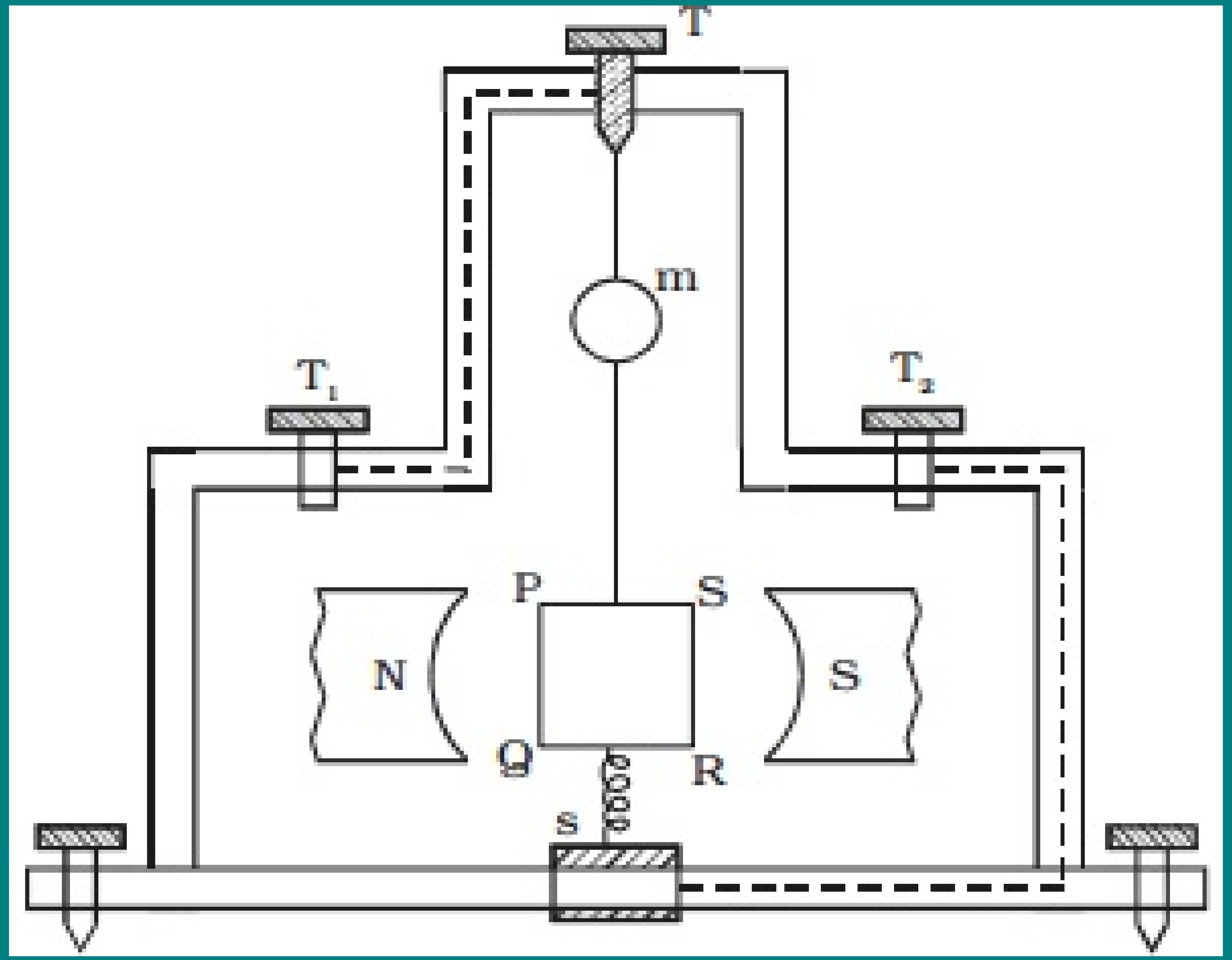

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- **When** a **current (I) flows** through the coil, a **torque acts on it**. This torque is given by $\boldsymbol{\tau = NIAB}$ ($\sin\theta = 1$ since the filed is radial.)
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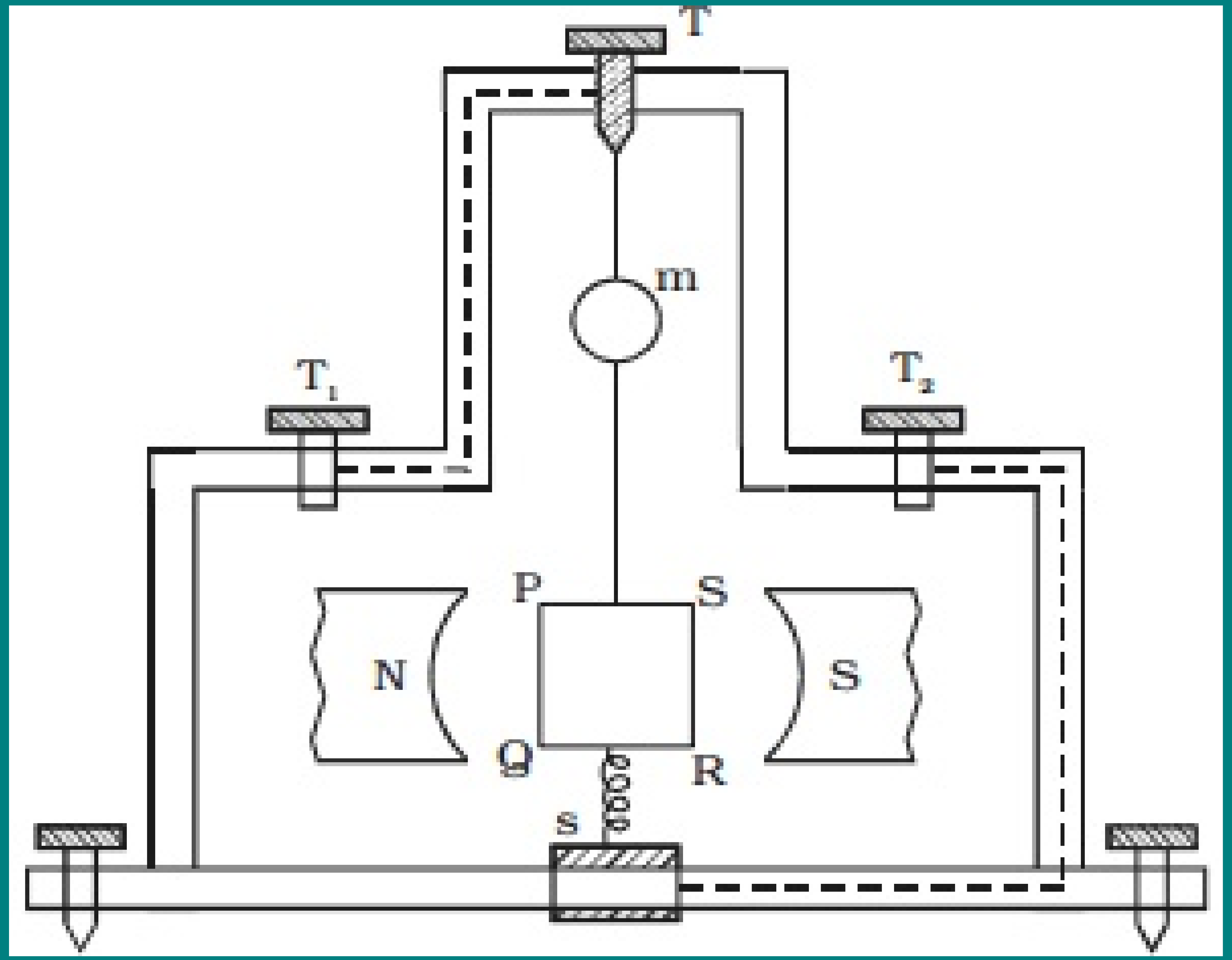

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- **When** a **current (I) flows** through the coil, a **torque acts on it**. This torque is given by $\boldsymbol{\tau = NIAB}$ ($\sin\theta = 1$ since the filed is radial.)
- This **magnetic torque** $(NIAB)$ **tends to rotate the coil**.
- The **spring provides** a **counter torque** $(\boldsymbol{k\phi})$ **that balances** the **magnetic torque** $(\boldsymbol{NIAB})$; resulting in a **steady angular deflection** (ϕ).

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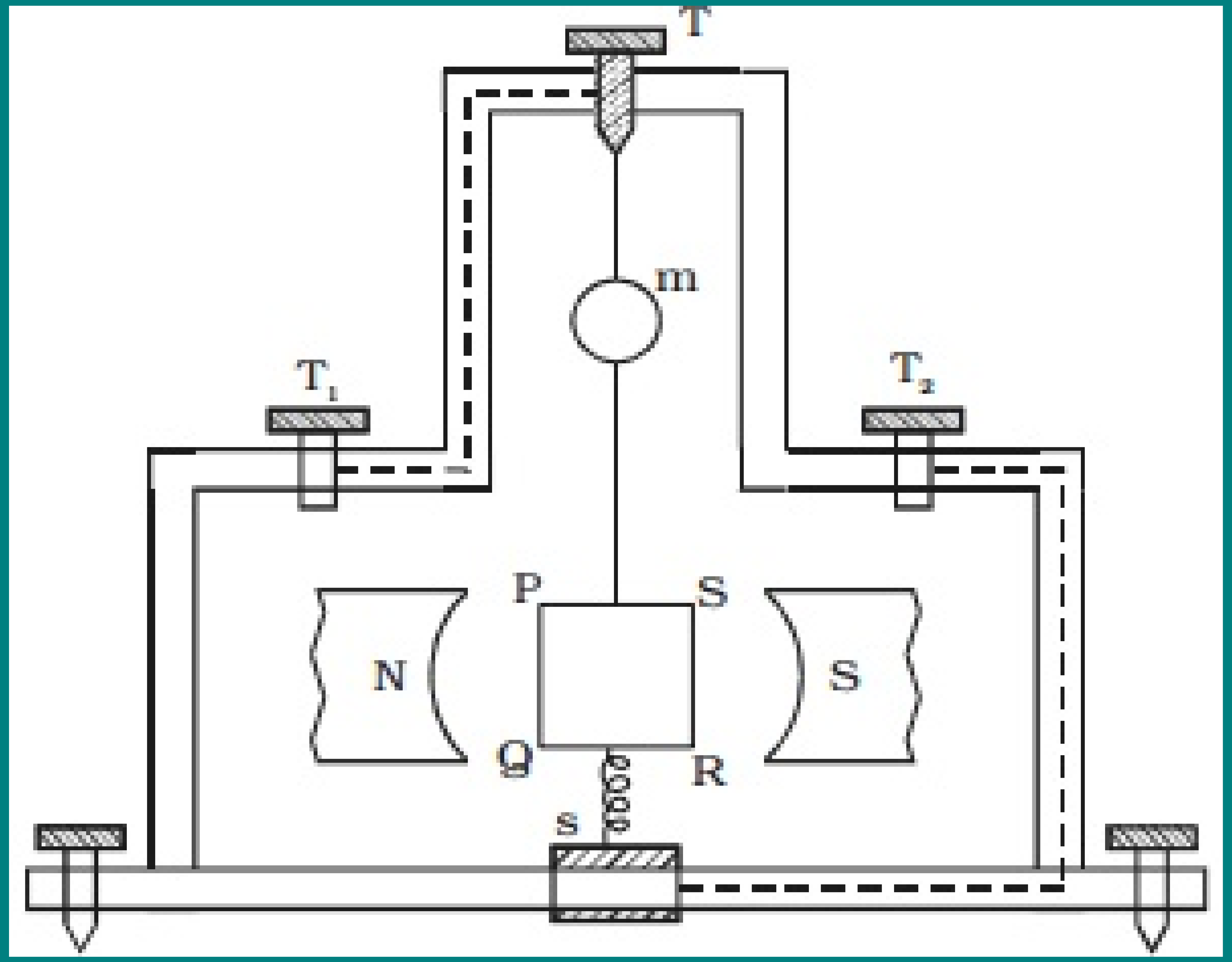

Working of A (MCG)

- **In equilibrium**

$$k\phi = NIAB$$

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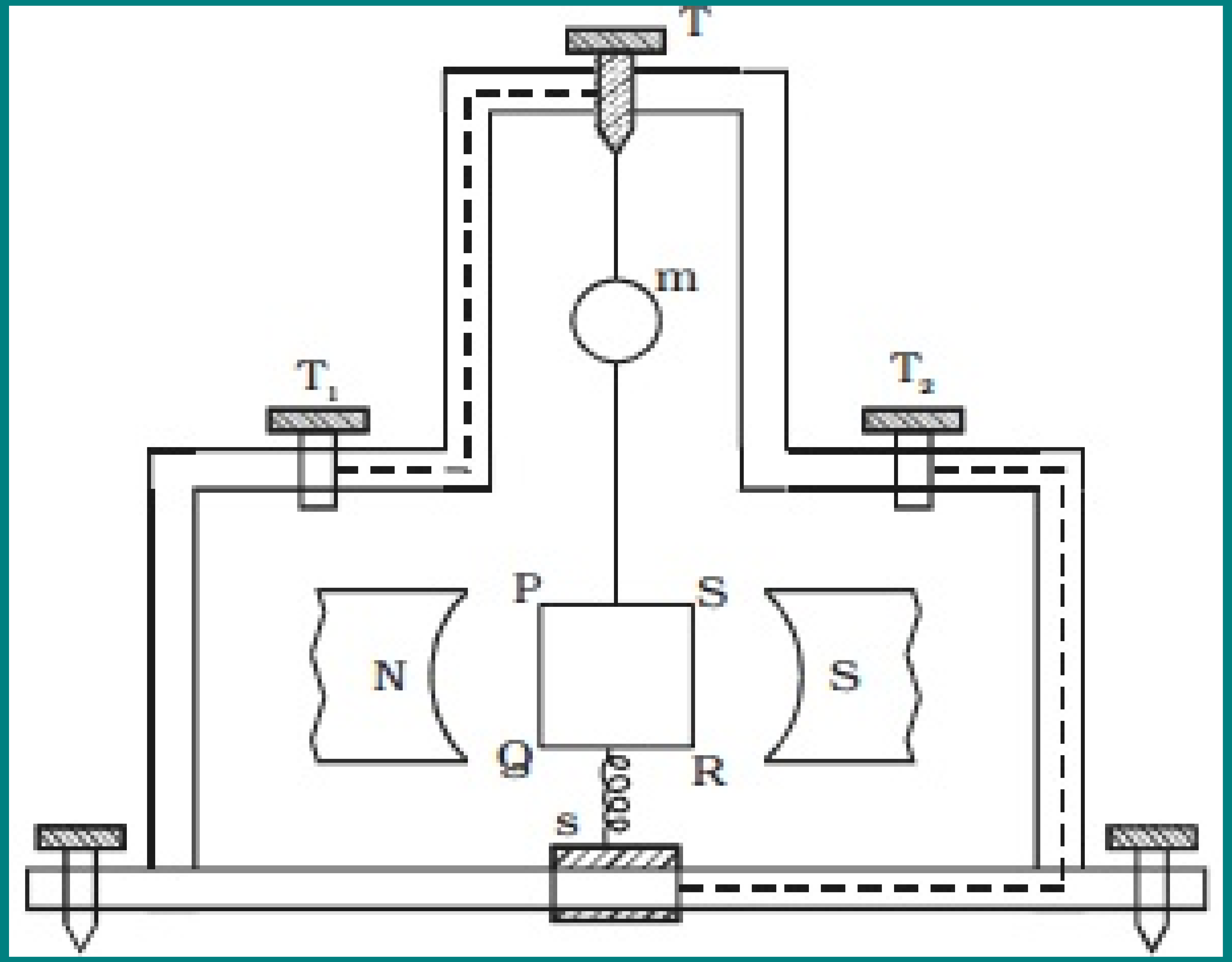

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- **In equilibrium**

$$k\phi = NIAB$$

$$\boxed{\phi = \frac{NIAB}{k}}$$

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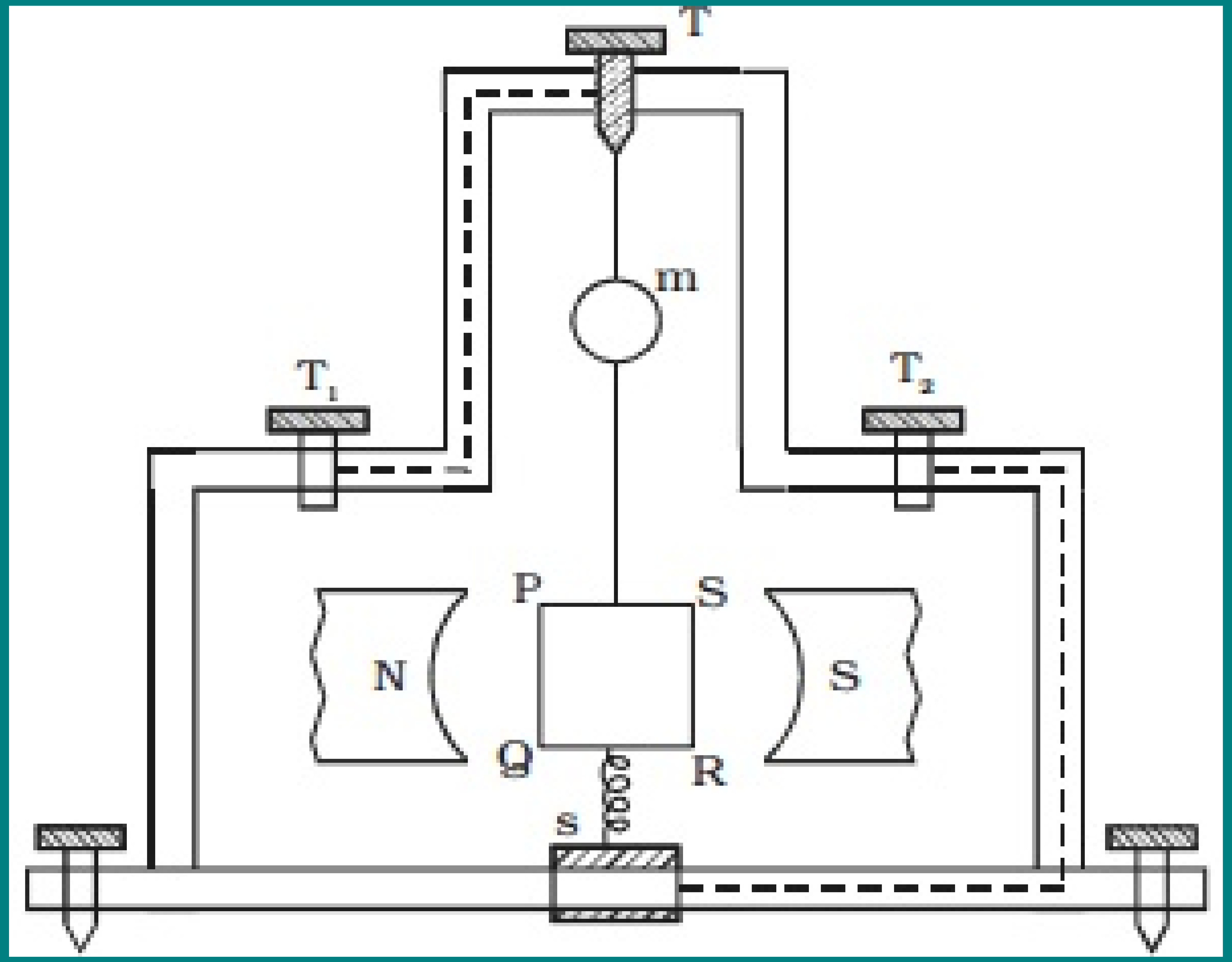

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- where, k is the **torsion constant of the spring**.

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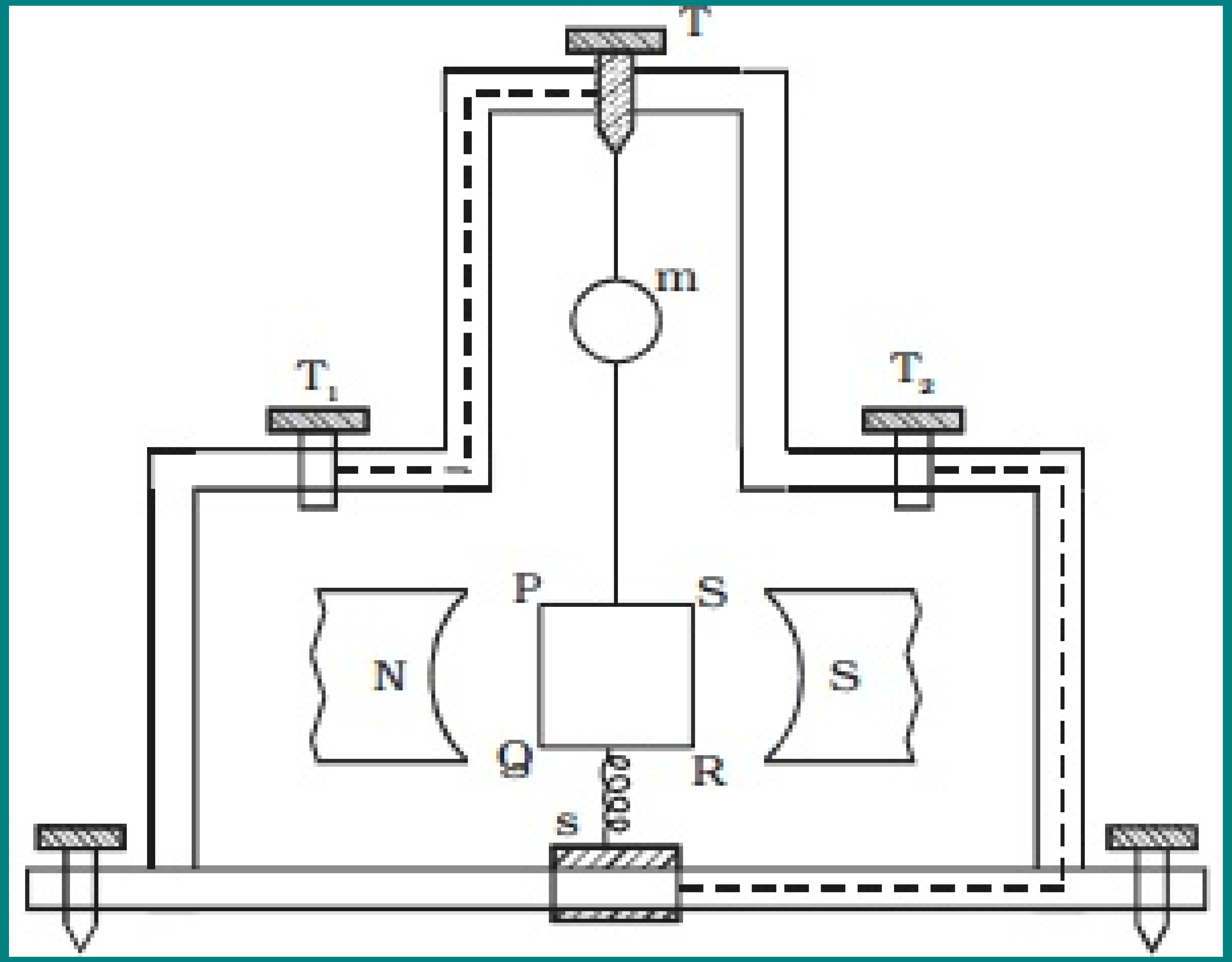

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$$k\phi = NIAB$$

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- where, k is the **torsion constant of the spring**.
- i.e $I \propto \phi$. Since the **deflection is directly proportional to the current** flowing through the coil, the **scale is linear** and is **calibrated to give directly the value of the current**.

Conversion of Galvanometer into an ammeter

THE MOVING COIL GALVANOMETER

Conversion of Galvanometer into an ammeter

- The **galvanometer cannot as such be used as an ammeter to measure** the value of the **current** in a given circuit. This is **for two reasons**:

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- **(ii) For measuring currents**, the **galvanometer has to be connected in series**, and as **it has a large resistance**, **this will change the value of the current** in the circuit.

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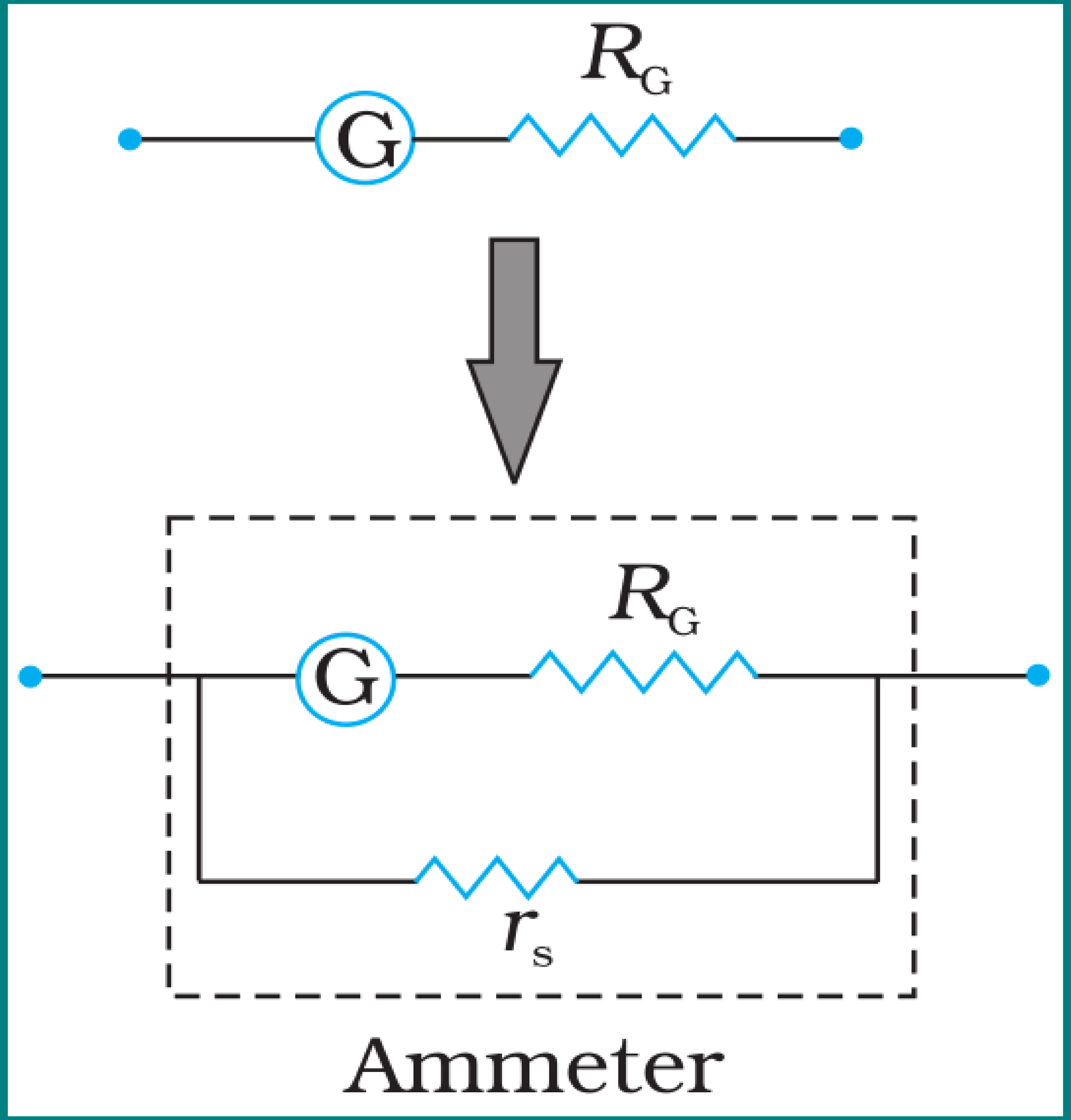
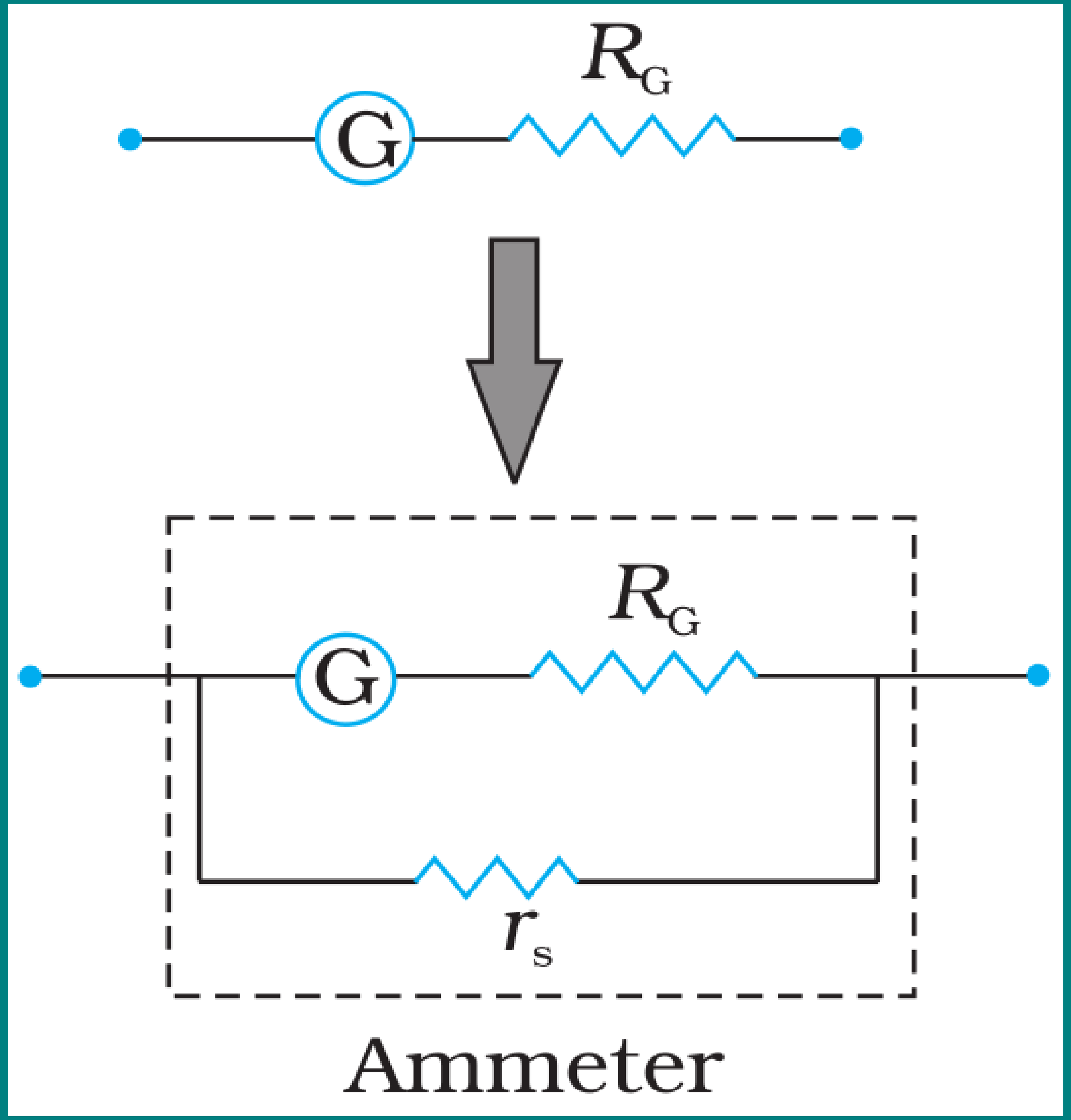


Conversion of Galvanometer into an ammeter

- However, a **galvanometer is converted into an ammeter by connecting a small resistance** (r_s), called **shunt resistance, in parallel with galvanometer**.

THE MOVING COIL GALVANOMETER



Conversion of Galvanometer into an ammeter

- However, a **galvanometer is converted into an ammeter by connecting a small resistance** (r_s), called **shunt resistance, in parallel with galvanometer**.
- As a result, **when large current flows** in a circuit, **only a small fraction of** the **current passes through** the **galvanometer** and the remaining **larger portion of** the **current passes through** the small **shunt resistance**.

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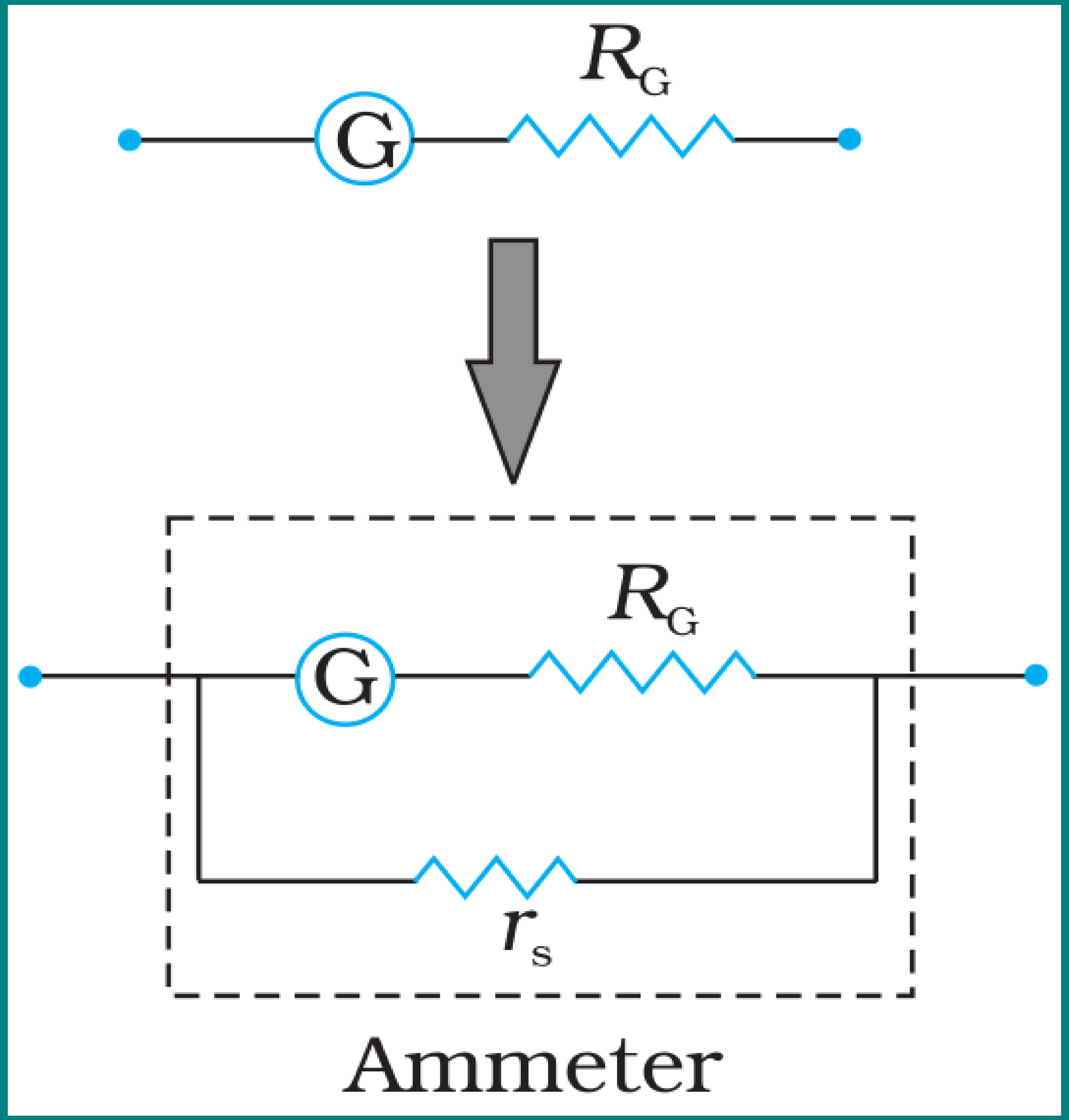

Conversion of Galvanometer into an ammeter

- The **effective resistance of** the **ammeter** R_a is

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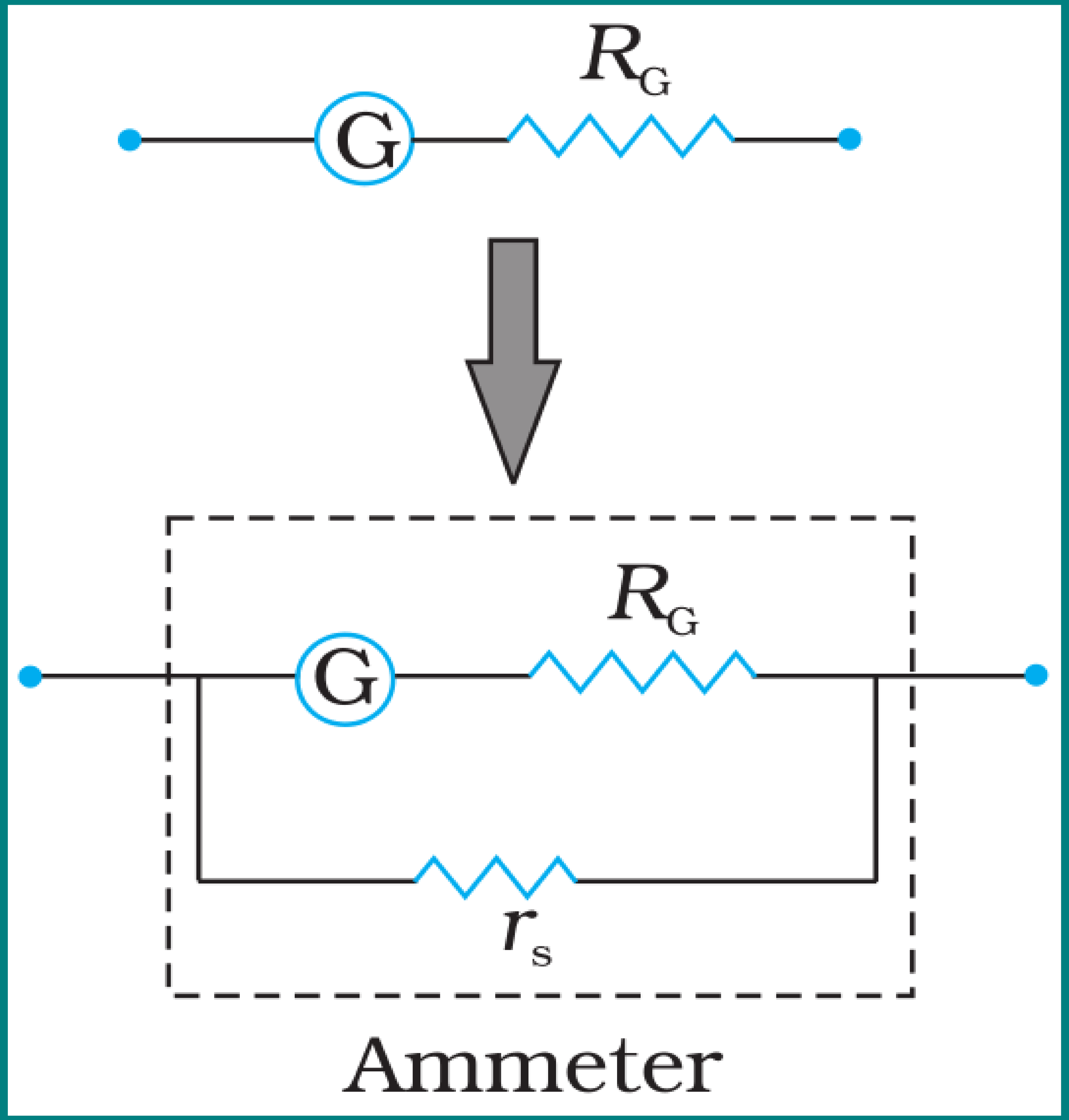

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$$\frac{1}{R_a} = \frac{1}{R_G} + \frac{1}{r_s} = \frac{r_s + R_G}{r_s\, R_G}$$

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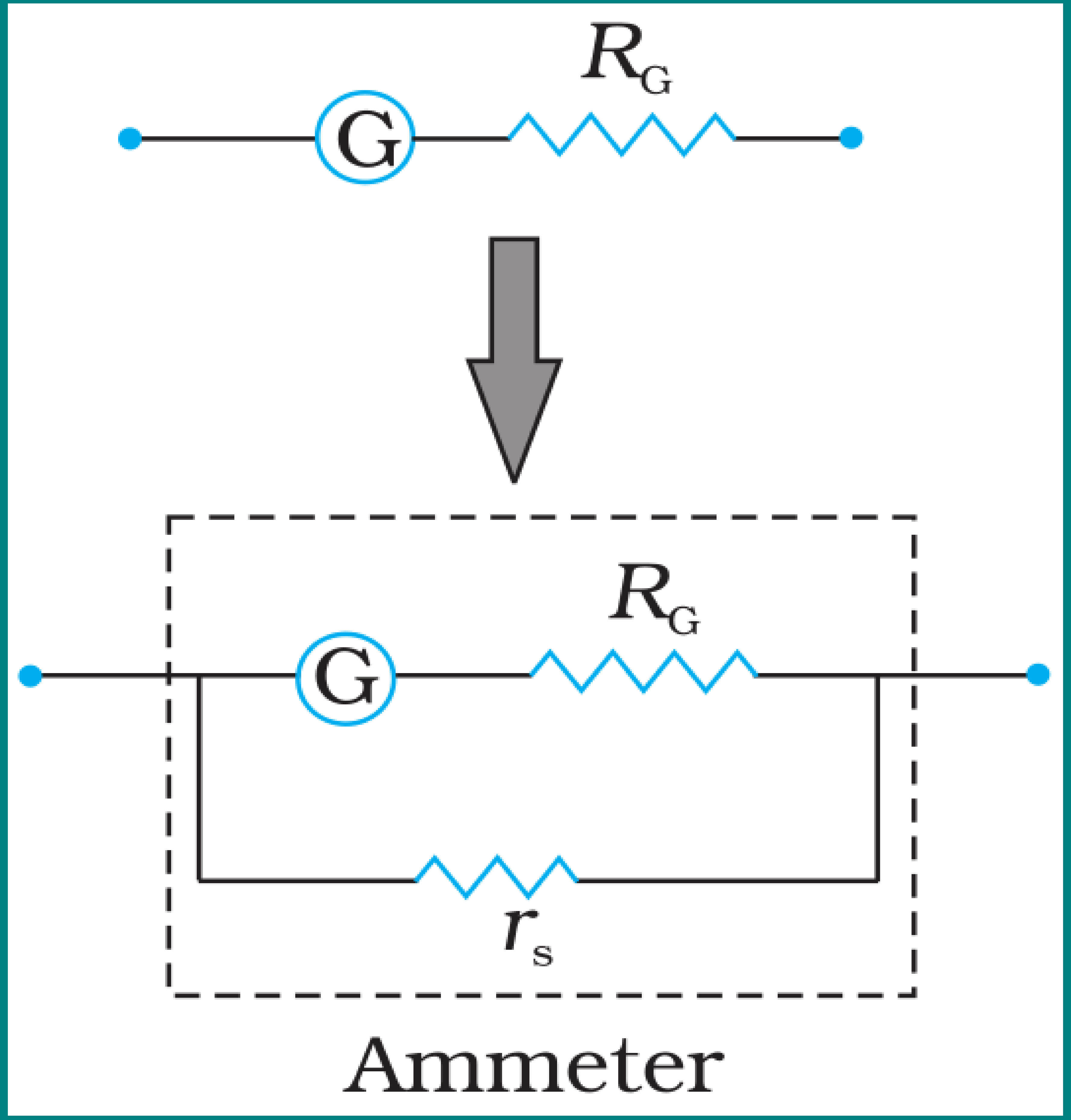

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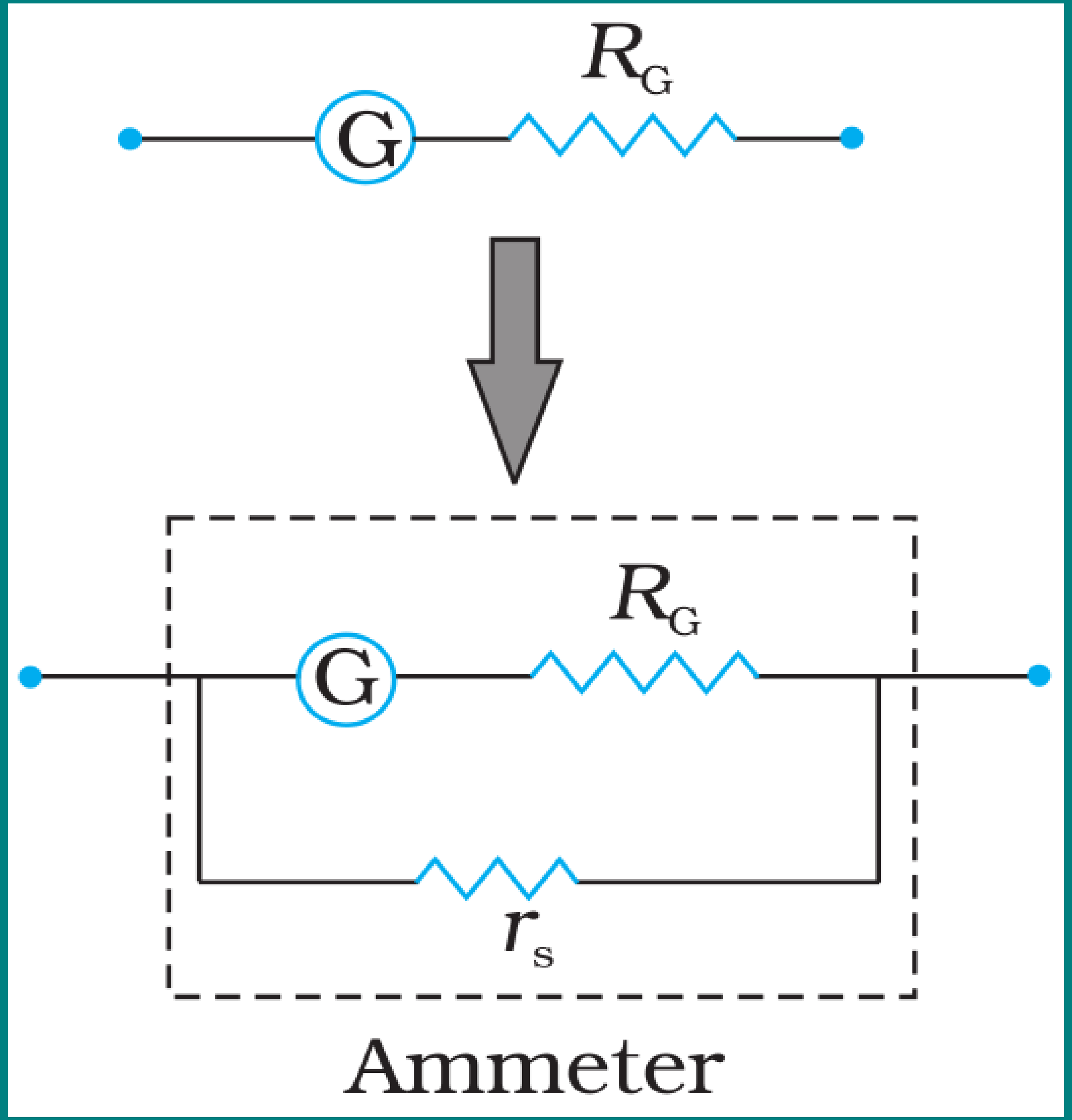

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- **If r_s** has **small** value, the **effect of** introducing the **measuring instrument is also small and negligible.**

THE MOVING COIL GALVANOMETER

Current sensitivity of a Galvanometer

- The **current sensitivity of** the **galvanometer** is **defined** as the **deflection** (ϕ) **per unit current** (I).

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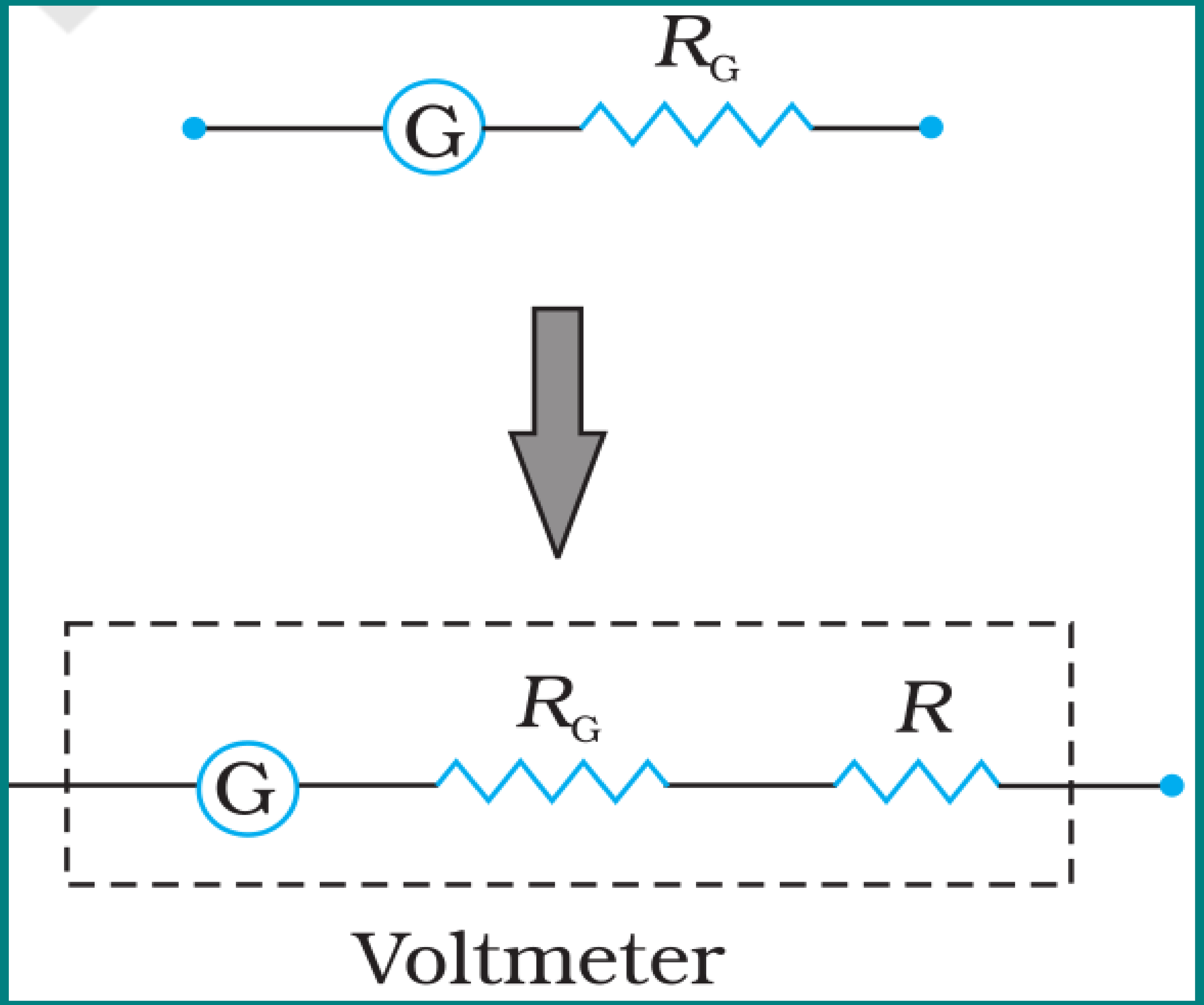

Conversion of Galvanometer into an voltmeter

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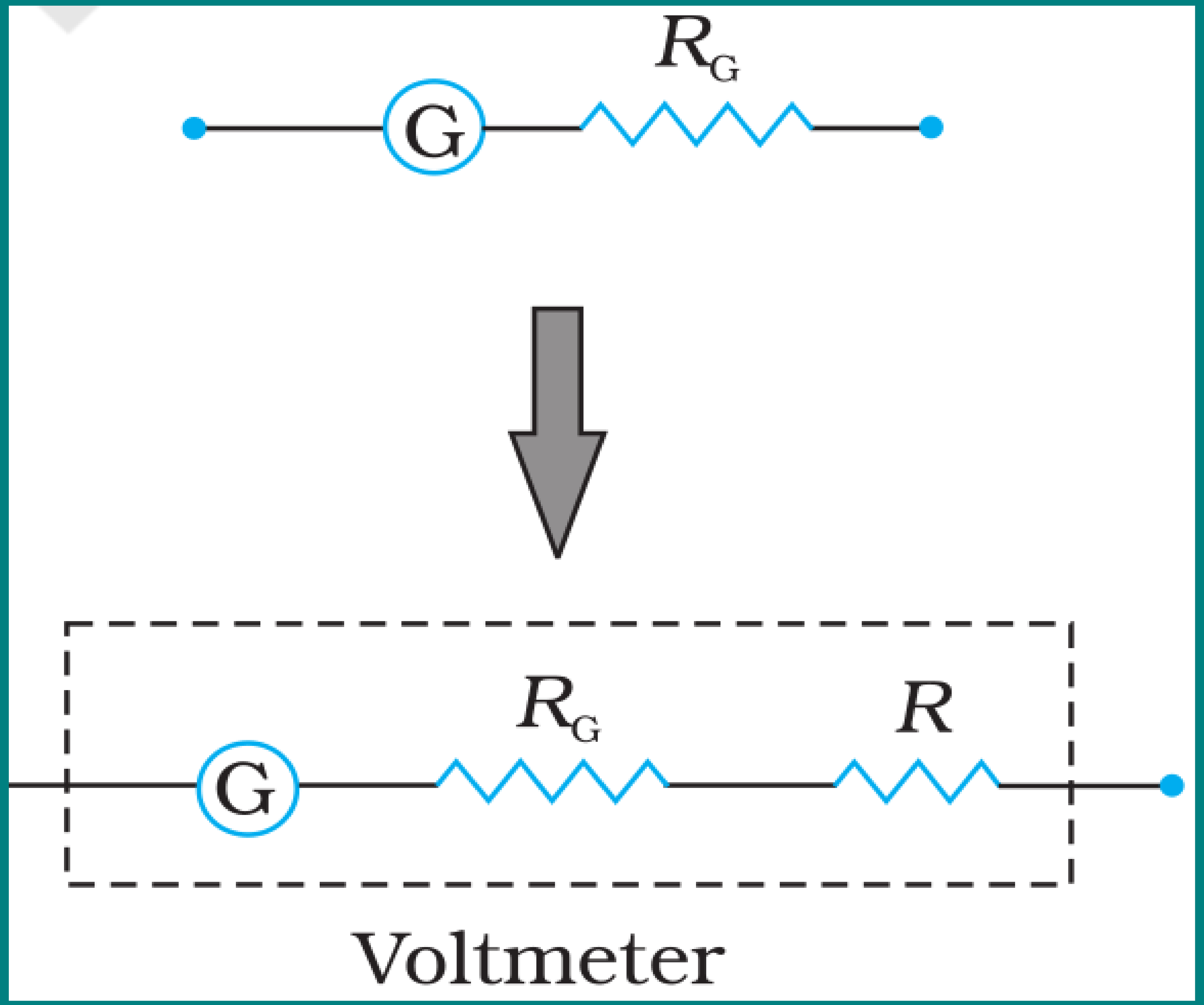

Conversion of Galvanometer into an voltmeter

- **Voltmeter** is an **instrument used to measure potential difference** between the two ends of a current carrying conductor. it must be **connected in parallel**.

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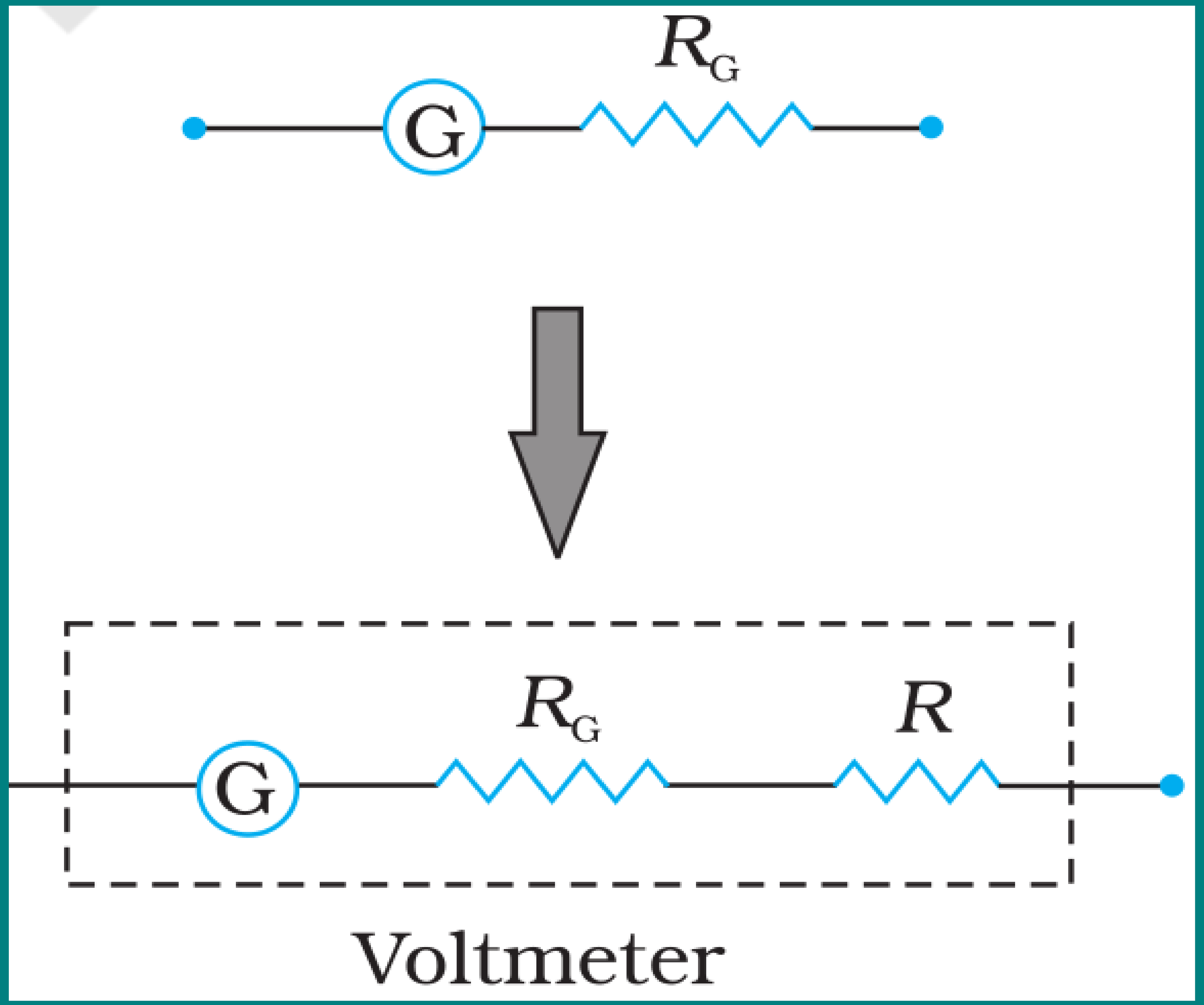

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- **Voltmeter** is an **instrument used to measure potential difference** between the two ends of a current carrying conductor. it must be **connected in parallel**.
- A **galvanometer** can be **converted into a voltmeter** by **connecting** a **high resistance (R) in series** with it.

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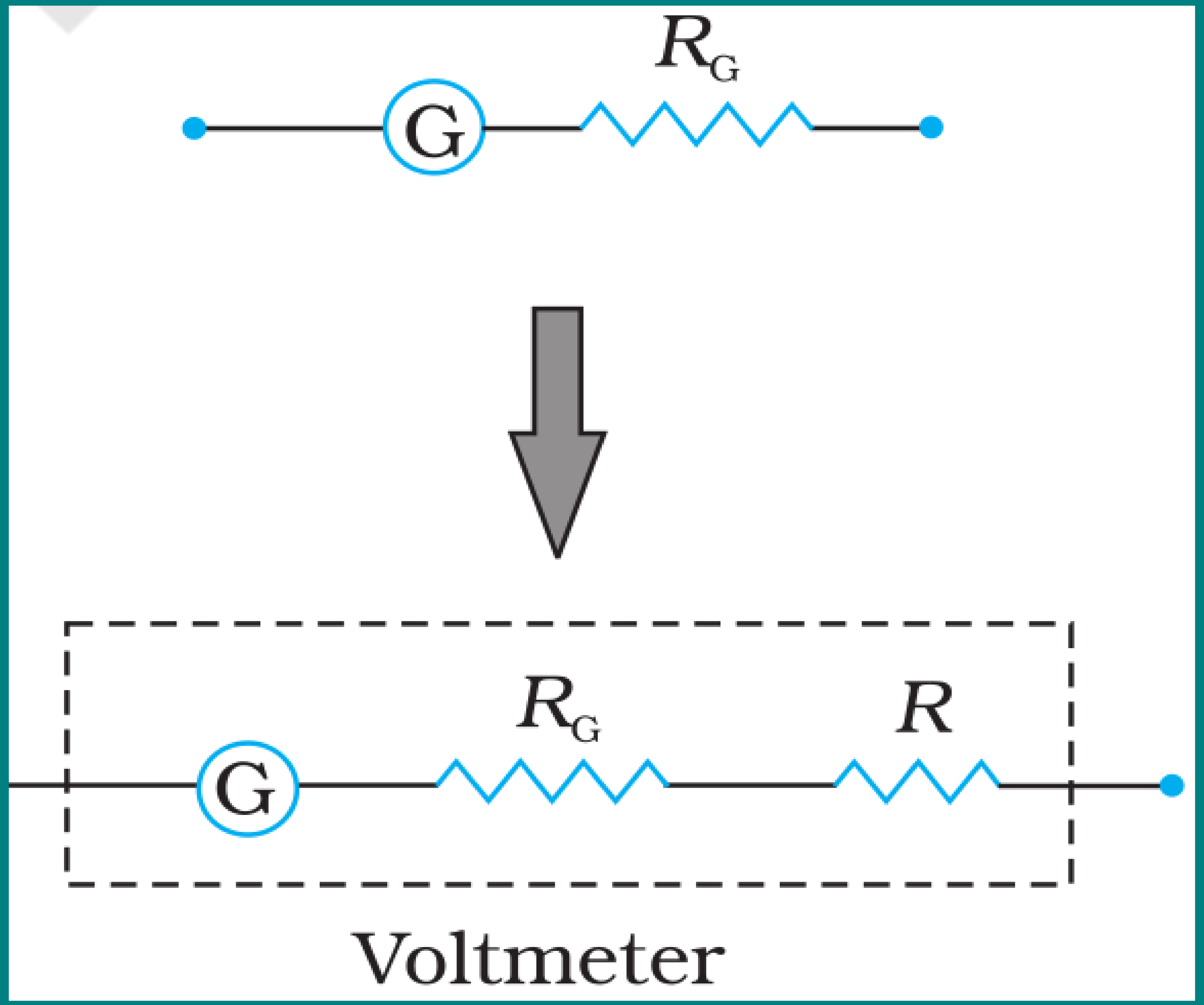

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- A **galvanometer** can be **converted into a voltmeter** by **connecting** a **high resistance (R) in series** with it.
- **Resistance of the voltmeter** is

$$R_v = R_G + R \approx R \text{ (Very large)}$$

THE MOVING COIL GALVANOMETER

Conversion of Galvanometer into an voltmeter

- R_v is **very large**, and hence **a voltmeter is connected in parallel** in a circuit as it **draws the least current from the circuit.** and the **measurement of potential difference is more accurate**.

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Voltage sensitivity of a Galvanometer

- The **voltage sensitivity of** the **galvanometer** is **defined** as the **deflection** (ϕ) **per unit voltage** (V).

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- The **voltage sensitivity of** the **galvanometer** is **defined** as the **deflection** (ϕ) **per unit voltage** (V).

$$\frac{\boldsymbol{\phi}}{\boldsymbol{V}} = \frac{\boldsymbol{NABI}}{\boldsymbol{kV}} = \frac{\boldsymbol{NAB}}{\boldsymbol{R}} \left(\because \frac{I}{V} = \frac{1}{R} \right)$$

THE MOVING COIL GALVANOMETER

Voltage sensitivity of a Galvanometer

- **Note: Increasing** the **current sensitivity may not necessarily increase** the **voltage sensitivity**.

THE MOVING COIL GALVANOMETER

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Voltage sensitivity of a Galvanometer

- **Note: Increasing** the **current sensitivity may not necessarily increase** the **voltage sensitivity**.
- The **current sensitivity** is given by : $\dfrac{\phi}{I} = \dfrac{NAB}{k}$
- **If we double the turns** $N \to 2N$, then **current sensitivity will also doubles** $\dfrac{\phi}{I} \to 2\dfrac{\phi}{I}$.

THE MOVING COIL GALVANOMETER

Voltage sensitivity of a Galvanometer

- **Note: Increasing** the **current sensitivity may not necessarily increase** the **voltage sensitivity**.
- The **current sensitivity** is given by : $\dfrac{\phi}{I} = \dfrac{NAB}{k}$
- **If we double the turns** $N \rightarrow 2N$, then **current sensitivity will also doubles** $\dfrac{\phi}{I} \rightarrow 2\dfrac{\phi}{I}$.
- But, **By doubling the turns** $N \rightarrow 2N$ the **resistance of the coil also doubles** $R \rightarrow 2R$, thus the **voltage sensitivity** $\dfrac{\phi}{V} \rightarrow \dfrac{\phi}{V}$ **remains unchanged**.

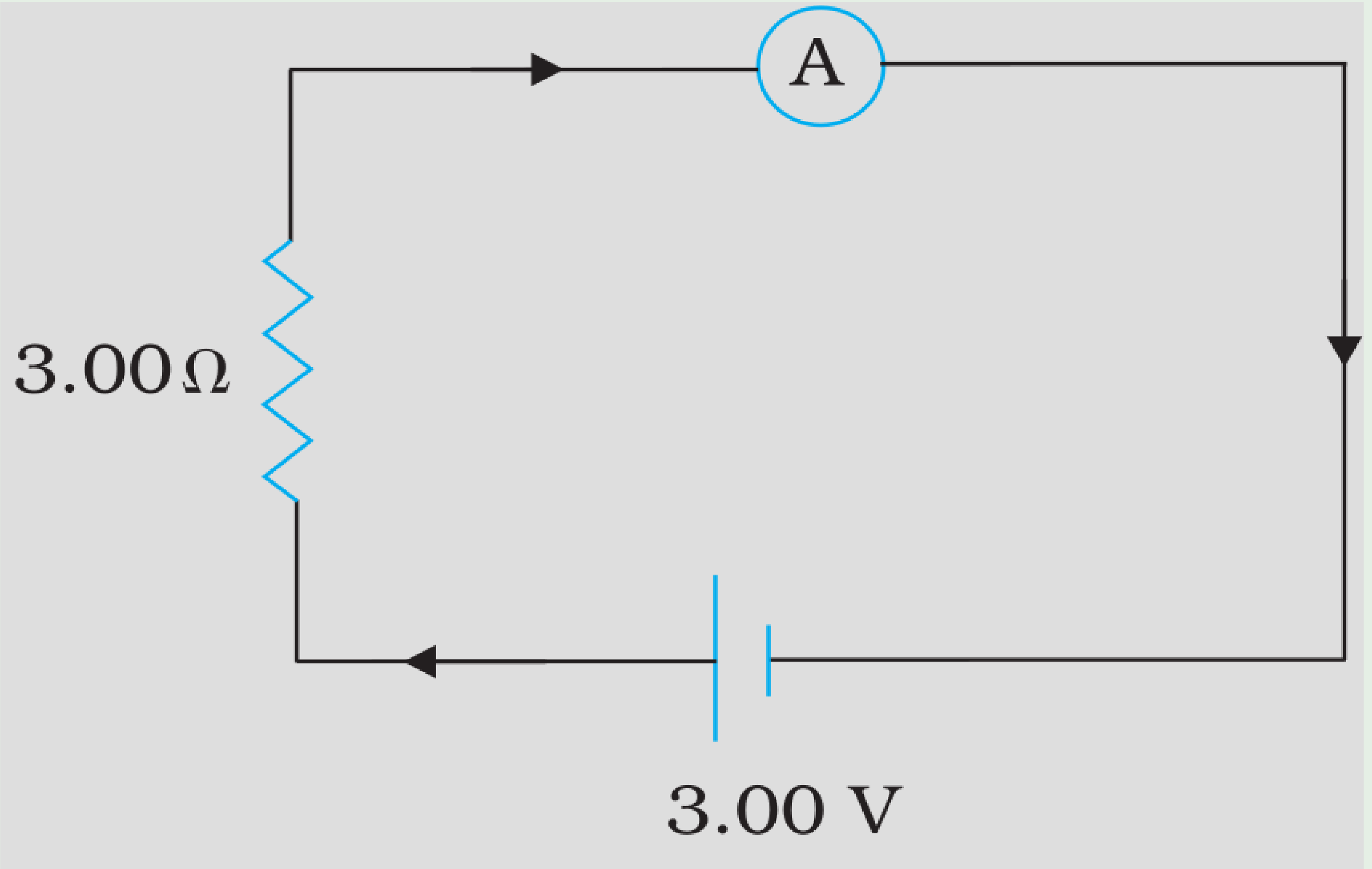
Example 4.13

In the circuit (Fig. 4.27) the current is to be measured. What is the value of the current if the ammeter shown

(a) is a galvanometer with a resistance $R_G = 60.00\ \Omega$;

(b) is a galvanometer described in (a) but converted to an ammeter by a shunt resistance $r_s = 0.02\ \Omega$;

(c) is an ideal ammeter with zero resistance?



Exercise 4.10

Two moving coil meters, M_1 and M_2 have the following particulars:
$R_1 = 10\ \Omega,\ N_1 = 30,\ A_1 = 3.6 \times 10^{-3}\ m^2,\ B_1 = 0.25\ T$
$R_2 = 14\Omega,\ N2 = 42,$
$A_2 = 1.8 \times 10^{-3}\ m^2, B_2 = 0.50\ T$
(The spring constants are identical for the two meters).
Determine the ratio of (a) current sensitivity and (b) voltage sensitivity of M_2 and M_1.

33. (a) (i) Write the principle and explain the working of a moving coil galvanometer. A galvanometer as such cannot be used to measure the current in a circuit. Why ?

(ii) Why is the magnetic field made radial in a moving coil galvanometer ? How is it achieved ? *5*

(i) For a moving coil galvanometer	
Principle	1
Working	1
Reason it cannot be used as such	1
(ii) Reason for radial field	1
How radial field is achieved	1

(i) Principle – When a rectangular loop carrying current I is placed in a uniform magnetic field, it experiences a torque. Working:- When a current flows through the coil of a galvanometer, a torque acts on it. $\tau = NiAB\sin\theta$ For radial magnetic field; $\sin\theta = 1$ The spring provides a counter or restoring toqrue $k\phi$. $k\phi = NiAB$ **In equilibrium;** $\phi = \left(\frac{NAB}{k}\right)i$	1 1
Galvanometer cannot be used as such to measure current because: -It has large resistance and hence will change the value of current in the circuit. -It is a sensitive device. (Any one of the above)	1
(ii) The magnetic field is made radial in a moving coil galvanometer so that the magnetic dipole moment ($\vec{m}$) is always perpendicular to the magnetic field ($\vec{B}$) Hence, $\sin\theta = 1$ always **Alternatively:** The magnetic field is made radial in a moving coil galvanometer to make the scale linear. It is achieved by using curved magnetic poles. **Alternatively:-**By using soft iron cylindrical core.	1 1

10. In a moving coil galvanometer, the deflecting torque τ acting on the coil is related to the current I flowing through it as :

(a) $\tau \propto I^3$

(b) $\tau \propto I^2$

(c) $\tau \propto I$

(d) $\tau \propto \sqrt{I}$

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(c) $\tau \ \alpha \ I$

24. Briefly explain why and how a galvanometer is converted into an ammeter. **2**

Explanation of conversion of galvanometer into an ammeter	
• Why	1
• How	1

24. Briefly explain why and how a galvanometer is converted into an ammeter. **2**

• Due to very high sensitivity **Alternatively** It has large resistance and hence will change the value of current in circuit.	1
• A galvanometer can be converted into an ammeter of desired range by connecting a shunt of proper value across its coil.	1

Explanation of conversion of galvanometer into an ammeter	
• Why	1
• How	1

29. (a) Briefly describe how the current sensitivity of a moving coil galvanometer can be increased.

(b) A galvanometer shows full scale deflection for current I_g. A resistance R_1 is required to convert it into a voltmeter of range $(0 - V)$ and a resistance R_2 to convert it into a voltmeter of range $(0 - 2V)$. Find the resistance of the galvanometer. **3**

(a)	Methods to increase current sensitivity of moving coil galvanometer	(1)
(b)	Calculation of resistance of galvanometer	(2)

(a) Current sensitivity of moving coil galvanometer can be increased by	
(1) increasing number of turns in the coil $(I_s \, \alpha \, N)$	
(2) increasing magnetic field strength $(I_s \, \alpha \, B)$	
(3) using a material having lesser value of torsional constant $\left(I_s \, \alpha \, \frac{1}{k}\right)$	
(4) increasing area of cross-section of the coil	
(**Note-** Any one of the above)	**1**
(b) $R = \frac{V}{I_g} - G$	½
For range $(0 - V)$	
$R_1 = \frac{V}{I_g} - G - - - - - - - - (1)$	½
For range (0-2V)	
$R_2 = \frac{2V}{I_g} - G - - - - - - - - (2)$	½
On solving equations (1) and (2)	½
$G = R_2 - 2R_1$	

29. (a) Briefly describe how the current sensitivity of a moving coil galvanometer can be increased.

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(a) Methods to increase current sensitivity of moving coil galvanometer	(1)
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(a) Current sensitivity of moving coil galvanometer can be increased by (1) increasing number of turns in the coil $(I_s \, \alpha \, N)$ (2) increasing magnetic field strength $(I_s \, \alpha \, B)$ (3) using a material having lesser value of torsional constant $\left(I_s \, \alpha \, \frac{1}{k}\right)$ (4) increasing area of cross-section of the coil (**Note-** Any one of the above)	**1**
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For range $(0 - V)$ $R_1 = \frac{V}{I_g} - G --------(1)$	½
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On solving equations (1) and (2) $G = R_2 - 2R_1$	½

29. (a) It is not advisable to use a galvanometer as such to measure current directly. Why ?

(b) Why should the value of resistance connected in parallel to a galvanometer be low ?

(c) Is the reading shown by an ammeter in a circuit less than or more than the actual value of current flowing in the circuit ? Why ? *3*

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(a) Explanation	1
(b) Explanation	1
(c) Answer	½
Explanation	½

(a) It will not measure accurate value of current because its high resistance will affect the current in the circuit.	1
(b) To reduce the galvanometer resistance a small resistance is connected in parallel.	1
(c) It is less than the actual value of current because it has some resistance	1

4. A galvanometer of resistance 100 Ω gives full scale deflection for a current of 1.0 mA. It is converted into an ammeter of range (0 – 1A). The resistance of the ammeter will be close to

(A) 0.1 Ω (B) 0.8 Ω

(C) 1.0 Ω (D) 10 Ω

4. A galvanometer of resistance 100 Ω gives full scale deflection for a current of 1.0 mA. It is converted into an ammeter of range (0 – 1A). The resistance of the ammeter will be close to

(A) 0.1 Ω (B) 0.8 Ω

(C) 1.0 Ω (D) 10 Ω

(A) 0.1Ω

1. An ammeter and a voltmeter are connected in series to a battery. Their readings are noted as ‘A’ and ‘V’ respectively. If a resistor is connected in parallel with the voltmeter, then

(A) A will increase, V will decrease.

(B) A will decrease, V will increase.

(C) Both A and V will decrease.

(D) Both A and V will increase.

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(C) Both A and V will decrease.

(D) Both A and V will increase.

(A) A will increase, V will decrease

4. A galvanometer shows full scale deflection for a current I_g. If a shunt of resistance S_1 is connected to the galvanometer, it gets converted into an ammeter of range $(0 - I)$. When resistance of the shunt is made S_2, its range becomes $(0 - 2I)$. Then $\left(\frac{S_1}{S_2}\right)$ is

(A) $\frac{I + I_g}{I - I_g}$

(B) $\frac{I - I_g}{I + I_g}$

(C) $\frac{2I - I_g}{I - I_g}$

(D) $\frac{I - I_g}{2I - I_g}$

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(D) $\frac{I - I_g}{2I - I_g}$

(C) $\frac{2I - I_g}{I - I_g}$

4. A galvanometer of resistance 50 Ω is converted into a voltmeter of range (0 – 2V) using a resistor of 1.0 kΩ. If it is to be converted into a voltmeter of range (0 – 10 V), the resistance required will be

(A) 4.8 kΩ

(B) 5.0 kΩ

(C) 5.2 kΩ

(D) 5.4 kΩ

4. A galvanometer of resistance 50 Ω is converted into a voltmeter of range (0 – 2V) using a resistor of 1.0 kΩ. If it is to be converted into a voltmeter of range (0 – 10 V), the resistance required will be

(A) 4.8 kΩ (B) 5.0 kΩ

(C) 5.2 kΩ (D) 5.4 kΩ

(C) $5.2 k\Omega$

8. A galvanometer of resistance 100 Ω is converted into an ammeter of range (0 – 1 A) using a resistance of 0.1 Ω. The ammeter will show full scale deflection for a current of about **1**

(A) 0.1 mA (B) 1 mA

(C) 10 mA (D) 0.1 A

8. A galvanometer of resistance 100 Ω is converted into an ammeter of range (0 – 1 A) using a resistance of 0.1 Ω. The ammeter will show full scale deflection for a current of about **1**

(A) 0.1 mA (B) 1 mA

(C) 10 mA (D) 0.1 A

(B) 1mA

(a) (i) (1) What is meant by current sensitivity of a galvanometer ? Mention the factors on which it depends.

(2) A galvanometer of resistance G is converted into a voltmeter of range (0 – V) by using a resistance R. Find the resistance, in terms of R and G, required to convert it into a voltmeter of range $\left(0 - \frac{V}{2}\right)$.

(i)(1) Meaning of current sensitivity, mentioning factors	2
(2) Finding the required resistance	1½
(ii) Finding the induced current	1½

(i) (1) Current sensitivity of galvanometer is defined as the deflection per unit current. **Alternatively**, $\frac{\phi}{I} = \frac{NBA}{K}$ **Factors** Number of turns in coil, Magnetic field intensity, Area of coil, Torsional Constant **(Any two)**	**1** ½+½
(2) $R = \frac{V}{I} - G$ for (0-V) Range $R_1 = \frac{V}{2I} - G$ for (0- $\frac{V}{2}$) Range $\frac{V}{I} = R + G$ $R_1 = \left(\frac{R+G}{2}\right) - G$	½ ½
$R_1 = \frac{R-G}{2}$	½

(b) (i) What is current sensitivity of a galvanometer ? Show how the current sensitivity of a galvanometer may be increased. “Increasing the current sensitivity of a galvanometer may not necessarily increase its voltage sensitivity.” Explain.

(ii) A moving coil galvanometer has a resistance 15 Ω and takes 20 mA to produce full scale deflection. How can this galvanometer be converted into a voltmeter of range 0 to 100 V ? *5*

(i) Definition of current sensitivity	1
Showing dependence of current sensitivity & explanation	1+1
(ii) Calculation of resistance	2

(i) Deflection produced per unit current is called its current sensitivity. $I_S = \frac{\theta}{I} = \frac{NBA}{K}$	**1**
Current sensitivity can be increased by (a) increasing number of turns in coil (b) increasing area of coil in magnetic field (c) decreasing K (Torsional Constant) **(any one)**	**1**
$V_s = \frac{\theta}{V} = \frac{NBA}{KR}$ If current sensitivity is increased by increasing number of turns of the coil, the resistance of the galvanometer will also increase. Thus voltage sensitivity may not increase.	**1**
(ii) $V = I_G(R+G)$ $R = \frac{V}{I_G} - G$	½
$= \frac{100}{20\times10^{-3}} - 15$	½
$= 5000 - 15$ $= 4985\Omega$	½
By connecting 4985Ω in series with galvanometer it is converted to voltmeter of range (0-100V)	½

5. A galvanometer of resistance G Ω is converted into an ammeter of range 0 to I A. If the current through the galvanometer is $0 \cdot 1\%$ of I A, the resistance of the ammeter is :

(A) $\frac{G}{999}\ \Omega$ (B) $\frac{G}{1000}\ \Omega$ (C) $\frac{G}{1001}\ \Omega$ (D) $\frac{G}{100 \cdot 1}\ \Omega$

5. A galvanometer of resistance G Ω is converted into an ammeter of range 0 to I A. If the current through the galvanometer is 0·1% of I A, the resistance of the ammeter is :

(A) $\frac{G}{999}\ \Omega$ (B) $\frac{G}{1000}\ \Omega$ (C) $\frac{G}{1001}\ \Omega$ (D) $\frac{G}{100 \cdot 1}\ \Omega$

(B) $\frac{G}{1000}\Omega$

12. The current sensitivity of a galvanometer does ***not*** depend on the :

(A) magnetic field in which the coil is suspended.

(B) current flowing in the coil.

(C) torsional constant of the spring.

(D) area of the coil.

12. The current sensitivity of a galvanometer does ***not*** depend on the :

(A) magnetic field in which the coil is suspended.

(B) current flowing in the coil.

(C) torsional constant of the spring.

(D) area of the coil.

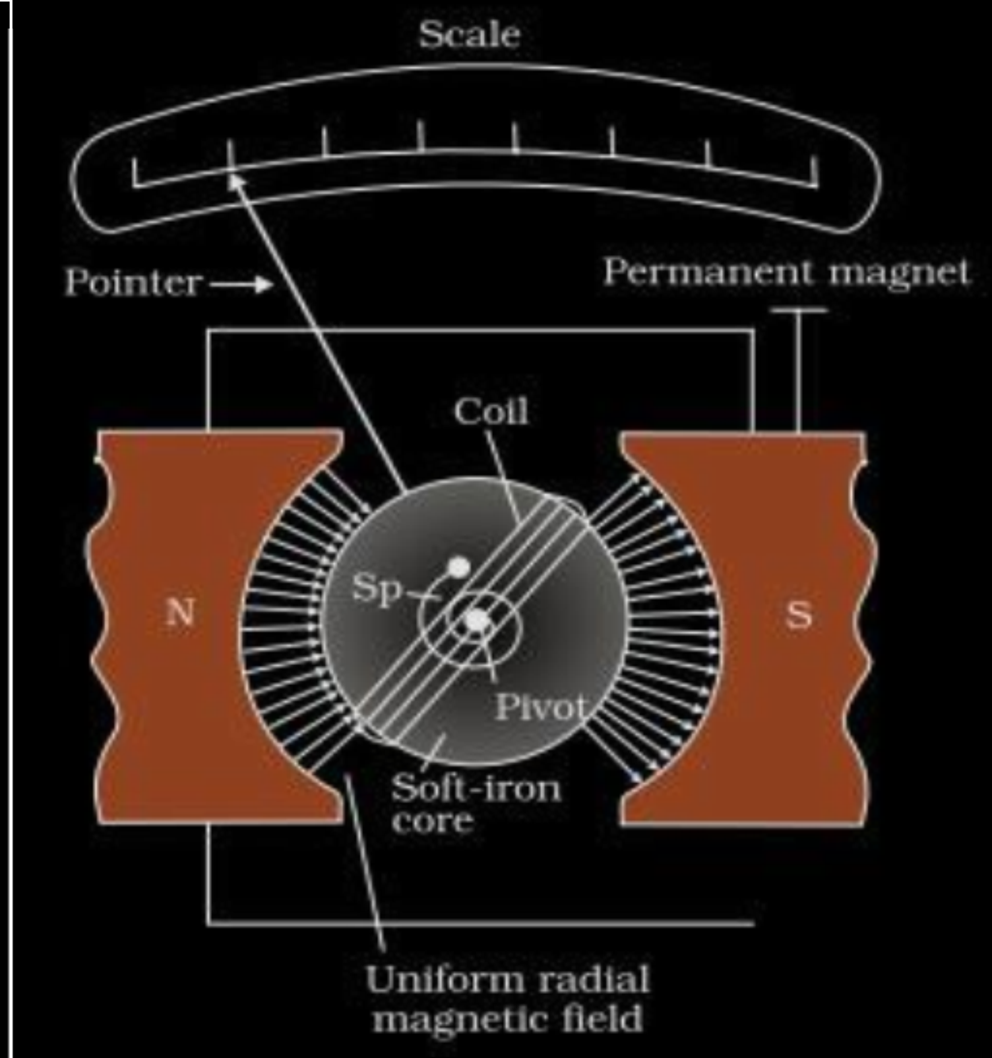
(B) Current flowing in the coil.

31. (a) (i) With the help of a labelled diagram, explain the principle of working of a moving coil galvanometer. Write the purpose of using (i) radial magnetic field, and (ii) soft iron core, in it.

(ii) Define current sensitivity of a galvanometer. "Increasing the current sensitivity may not necessarily increase the voltage sensitivity." Give reason. *5*

(i) Labelled diagram	1
Working principle of moving coil galvanometer	1
Use of (i) Radial magnetic field	½
(ii) Soft iron core	½
(ii) Defining current sensitivity	1
Reason	1

Principle:A current carrying coil placed in uniform magnetic field experiences a torque.	1
(i) Radial magnetic field makes the scale linear Alternatively: Radial magnetic field provides maximum Torque.	½
(ii) Use of soft iron core is to increase the strength of magnetic field/ increase sensitivity of the galvanometer.	½
(ii) **Current sensitivity** is defined as deflection per unit current. **<u>Alternatively</u>**: $\mathrm{I}_s = \frac{\Phi}{\mathrm{I}} = \frac{NAB}{k}$	1
Voltage sensitivity $V_s = \frac{\Phi}{V} = \left(\frac{NAB}{k}\right)\frac{\mathrm{I}}{V} = \left(\frac{NAB}{k}\right)\frac{\mathrm{I}}{R}$	½
Increase in number of turns, increases the current sensitivity and resistance of the galvanometer in the same proportion of current sensitivity therefore Voltage sensitivity remains unchanged.	½



1

(b) (i) (I) Write Ampere's circuital law in mathematical form and explain the terms used.

(II) As the current carrying solenoid is made longer, the magnetic field produced outside it approaches zero. Why ?

(III) A flexible loop of irregular shape carrying current when located in an external magnetic field, changes to a circular shape. Give reason.

(ii) A galvanometer of resistance G is converted into a voltmeter to measure up to V volts, by connecting a resistance R_1 in series with the coil. If R_1 is replaced by R_2, then it can only measure up to $\frac{V}{2}$ volt. Find the value of the resistance R_3 (in terms of R_1 and R_2) needed to convert it into a voltmeter that can read up to 2 V. *5*

(i) (I) Writing Ampere circuital law & explaining the terms.	1
(II) Reason for magnetic field outside long solenoid approaching zero	1
(III) Reason for irregular shaped loop changing to circular loop in uniform magnetic field	1
(ii) Finding the value of Resistance R_3	2

(i) **(I)** $\oint \vec{B}.\vec{dl}=\mu_0 I_e$	½
I_e= Total current through the surface B= Magnetic field dl = length of small element	½
(II) As length of solenoid increases, it appears like a long cylindrical metal sheet so field outside approaches zero.	1
(III) For a given perimeter, a circle encloses greater area than any other shape, which maximizes the flux.	1
(ii) $R_1=\frac{V}{I_g}-G \qquad \Rightarrow \frac{V}{I_g}=R_1+G \qquad$ -----(1)	½
$R_2=\frac{V}{2I_g}-G \qquad \Rightarrow \frac{V}{2I_g}=R_2+G \qquad$ -----(2)	½
Solving (1) & (2) $G=R_1-2R_2$	½
$R_3=\frac{2V}{I_g}-G \qquad$ -----(3)	
Solving using eq (1) & (3) $R_3=3R_1-2R_2$	½

11. A galvanometer can be converted into an ammeter of desired range by connecting a :

(A) small resistance in series

(B) large resistance in series

(C) small resistance in parallel

(D) large resistance in parallel

11. A galvanometer can be converted into an ammeter of desired range by connecting a :

(A) small resistance in series (B) large resistance in series

(C) small resistance in parallel (D) large resistance in parallel

(C) small resistance in parallel

29. A galvanometer is an instrument used to show the direction and strength of the current passing through it. In a galvanometer, a coil placed in a magnetic field experiences a torque and hence gets deflected when a current passes through it. The name is derived from the surname of Italian scientist L. Galvani, who in 1791 discovered that electric current makes a dead frog's leg jerk. A spring attached with the coil provides a counter torque.

In equilibrium, the deflecting torque is balanced by the restoring torque of the spring and we have :

$$NBAI = k\phi$$

where N is the total number of turns in the coil

A is the area of cross-section of each turn

B is the radial magnetic field

k is the torsional constant of the spring

ϕ is the angular deflection of the coil

As the current (I_g) which produces full scale deflection in the galvanometer is very small, the galvanometer cannot as such be used to measure current in electric circuits. A small resistance, called shunt, of a suitable value is connected with the galvanometer to convert it into an ammeter of desired range. By using a higher resistance, a galvanometer can also be converted into a voltmeter.

(i) (a) A galvanometer is converted into a voltmeter of range (0 – V) by connecting with it, a resistance R_1. If R_1 is replaced by R_2, the range becomes (0 – 2 V). The resistance of the galvanometer is : *1*

(A) $(R_2 - 2R_1)$ (B) $(R_2 - R_1)$

(C) $(R_1 + R_2)$ (D) $(R_1 - 2R_2)$

(i) (a) A galvanometer is converted into a voltmeter of range $(0 - V)$ by connecting with it, a resistance R_1. If R_1 is replaced by R_2, the range becomes $(0 - 2\ V)$. The resistance of the galvanometer is : *1*

(A) $(R_2 - 2R_1)$ (B) $(R_2 - R_1)$

(C) $(R_1 + R_2)$ (D) $(R_1 - 2R_2)$

(A) $(R_2 - 2R_1)$

(b) A current of 5 mA flows through a galvanometer. Its coil has 100 turns, each of area of cross-section $18\ \text{cm}^2$ and is suspended in a magnetic field 0·20 T. The deflecting torque acting on the coil will be : *1*

(A) $3{\cdot}6 \times 10^{-3}$ Nm (B) $1{\cdot}8 \times 10^{-4}$ Nm

(C) $2{\cdot}4 \times 10^{-3}$ Nm (D) $1{\cdot}2 \times 10^{-4}$ Nm

(B) 1.8×10^{-4} Nm

(ii) The value of resistance of the ammeter in case (ii) will be : *1*

(A) $0{\cdot}20\ \Omega$ (B) $0{\cdot}24\ \Omega$

(C) $6{\cdot}0\ \Omega$ (D) $6{\cdot}25\ \Omega$

Award 1 mark for this question to all students .

(iii) A galvanometer of resistance 6 Ω shows full scale deflection for a current of 0·2 A. The value of shunt to be used with this galvanometer to convert it into an ammeter of range (0 – 5 A) is : *1*

(A) 0·25 Ω (B) 0·30 Ω

(C) 0·50 Ω (D) 6·0 Ω

(iii) A galvanometer of resistance 6 Ω shows full scale deflection for a current of 0·2 A. The value of shunt to be used with this galvanometer to convert it into an ammeter of range (0 – 5 A) is : *1*

(A) 0·25 Ω (B) 0·30 Ω

(C) 0·50 Ω (D) 6·0 Ω

(A) 0.25 Ω

(iv) The value of the current sensitivity of a galvanometer is given by : *1*

(A) $\frac{k}{NBA}$ (B) $\frac{NBA}{k}$

(C) $\frac{kBA}{N}$ (D) $\frac{kNB}{A}$

(iv) The value of the current sensitivity of a galvanometer is given by : *1*

(A) $\dfrac{k}{NBA}$ (B) $\dfrac{NBA}{k}$

(C) $\dfrac{kBA}{N}$ (D) $\dfrac{kNB}{A}$

(B) $\dfrac{NBA}{K}$

19. A voltmeter of resistance 1000 Ω can measure up to 25 V. How will you convert it so that it can read up to 250 V ? *2*

19. A voltmeter of resistance 1000 Ω can measure up to 25 V. How will you convert it so that it can read up to 250 V ? *2*

Conversion of voltmeter to read upto 250V 2	
$I = \frac{V}{R}$	
$= \frac{25}{1000}$	½
$= 25\times10^{-3}$ A	
Resistance to be connected to voltmeter	
$R' = \frac{V'}{I} - R$	½
$= \frac{250}{25\times10^{-3}} - 1000$	
$= 9000\,\Omega$	½
This 9000 Ω is in series with voltmeter.	½

16. **Assertion (A)** : The deflection in a galvanometer is directly proportional to the current passing through it. **1**

Reason (R) : The coil of a galvanometer is suspended in a uniform radial magnetic field.

16. **Assertion (A)** : The deflection in a galvanometer is directly proportional to the current passing through it. **1**

Reason (R) : The coil of a galvanometer is suspended in a uniform radial magnetic field.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

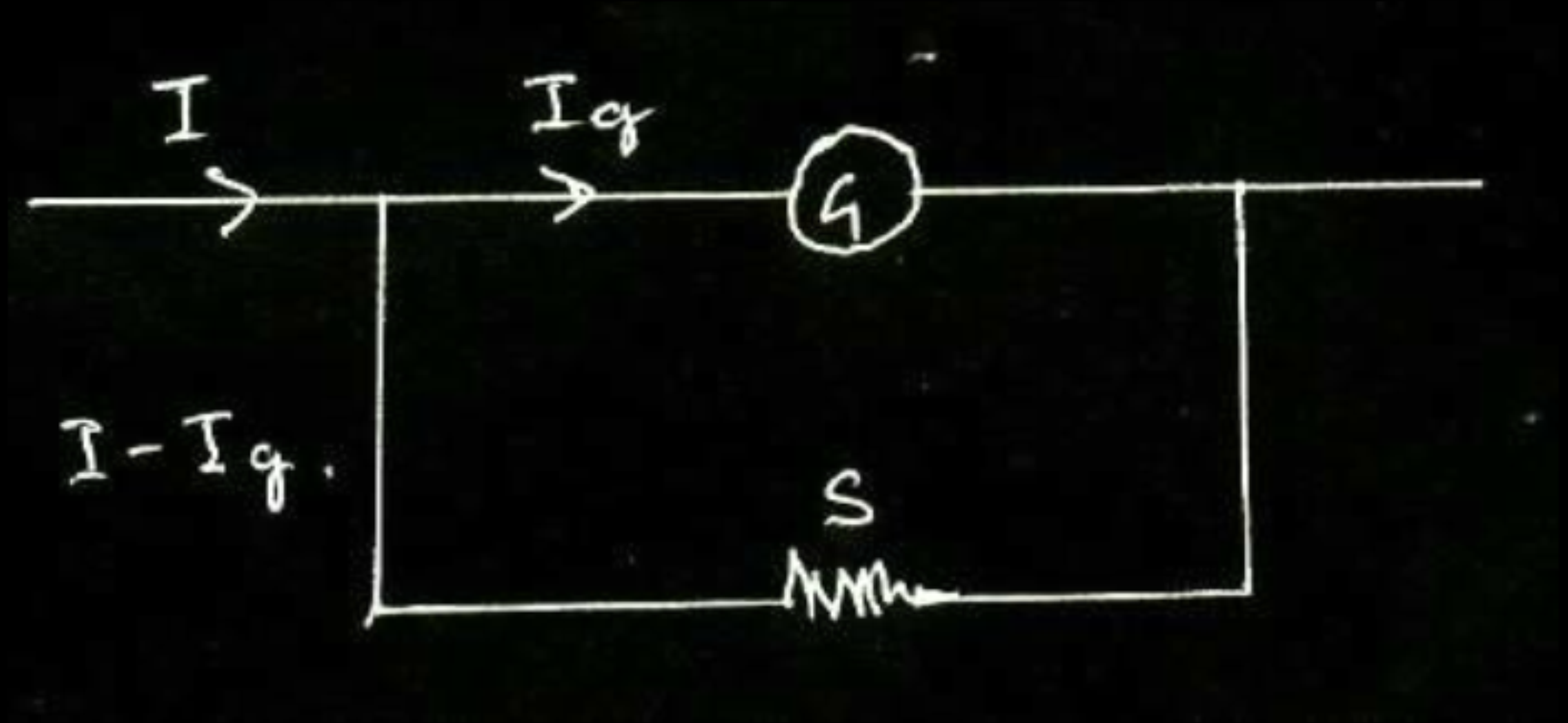
(b) To convert a given galvanometer into a voltmeter of ranges 2 V, V and $\frac{V}{2}$ volt, resistances R_1, R_2 and R_3 ohm respectively, are required to be connected in series with the galvanometer. Obtain the relationship between R_1, R_2 and R_3.

b)	
$$R = \frac{V}{i_g} - G$$	½
$$R_1 = \frac{2V}{i_g} - G = R_o - G$$	
$$R_1 + G = 2R_o$$	½
$$\left[Where\ R_o = \frac{V}{i_g}\right]$$	
Similarily	
$$R_2 + G = R_o$$	
$$R_3 + G = R_o/2$$	½
From the above equations,	
$$R_1 - R_2 = 2(R_2 - R_3$$	
$$R_1 - 3R_2 + 2R_3 = 0$$	½

22. An ammeter of resistance 0·8 Ω can measure a current up to 1·0 A. Find the value of shunt resistance required to convert this ammeter to measure a current up to 5·0 A. *2*

22. An ammeter of resistance 0·8 Ω can measure a current up to 1·0 A. Find the value of shunt resistance required to convert this ammeter to measure a current up to 5·0 A. *2*

Diagram	½
Formula	½
Calculation of value of shunt	1



½

$$I_g = 1\ A$$

Resistance of ammeter, $R_A = 0.8\ \Omega$

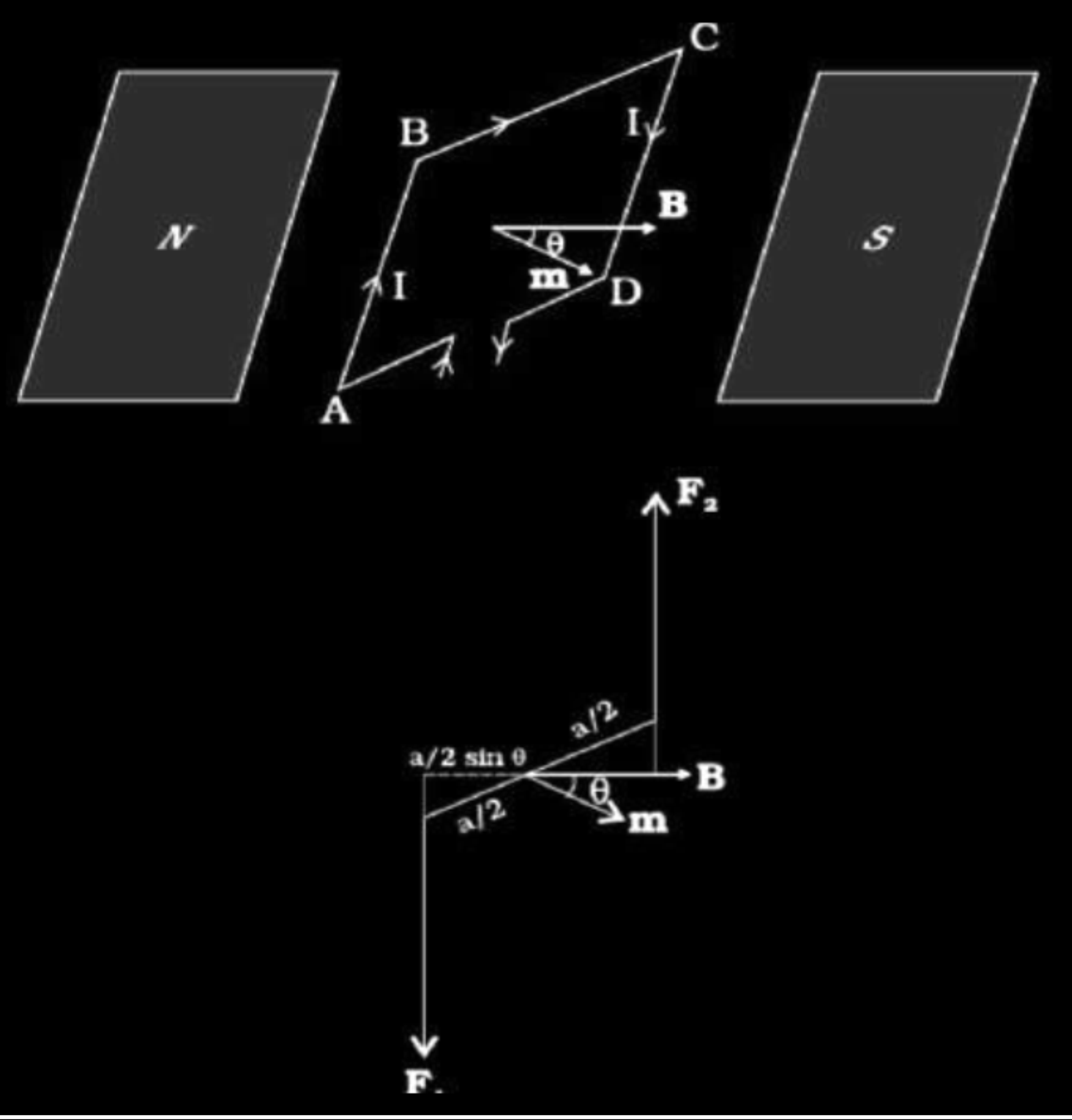
$$I_g R_A = (I - I_g)\, S$$ 1

$$\Rightarrow 1 \times 0.8 = (5 - 1)\, S$$

$$\Rightarrow \quad S = 0.2\ \Omega$$ ½

(a) Obtain the expression for the deflecting torque acting on the current carrying rectangular coil of a galvanometer in a uniform magnetic field. Why is a radial magnetic field employed in the moving coil galvanometer ?

(a)



½

Arms AD and BC experience no net force whereas arm AB and CD experience forces which constitute torque

$F_1 = F_2 = IbB$ ½

Therefore magnitude of torque

$$\tau = \left(F_1 \frac{a}{2} + F_2 \frac{a}{2} \right) \sin\theta$$ ½

$$\tau = I(ab)\sin\theta$$

$$where\ ab = A(area\ of\ the\ loop)$$

$$\Rightarrow \tau = IAB\sin\theta$$ ½

$$for\ \ N\ number\ of\ turns$$

$$\tau = NIAB\sin\theta$$

$$\tau = \vec{M} \times \vec{B}$$

Where magnetic moment M=NIA

Galvanometer has Radial magnetic field to increase the field strength and to make torque independent of orientation θ/ it maximise the torque 1

(a) Define current sensitivity of a galvanometer. Write its expression.

(b) A galvanometer has resistance G and shows full scale deflection for current I_g.

(i) How can it be converted into an ammeter to measure current up to I_0 $(I_0 > I_g)$?

(ii) What is the effective resistance of this ammeter ? *3*

a) Definition and expression	1 mark
b) Conversion of Galvanometer	
(i) into ammeter	1 mark
(ii) Effective resistance	1 mark

a) Deflection per unit current	½
$$I_s = \frac{\theta}{I} = \frac{BNA}{K}$$	½
b) (i) By connecting a low resistance (R_s) in parallel to galvanometer such that	½
$(I_0 - I_g)R_s = I_g G$	½
(ii) effective resistance $$\frac{1}{R_A} = \frac{1}{R_s} + \frac{1}{G} = \frac{G + R_s}{R_s G}$$	
$$\therefore R_A = \frac{R_s G}{G + R_s}$$	1

State the principle of a moving coil galvanometer. Explain its working and obtain the expression for the deflection produced due to the current passed through the coil. Define current sensitivity.

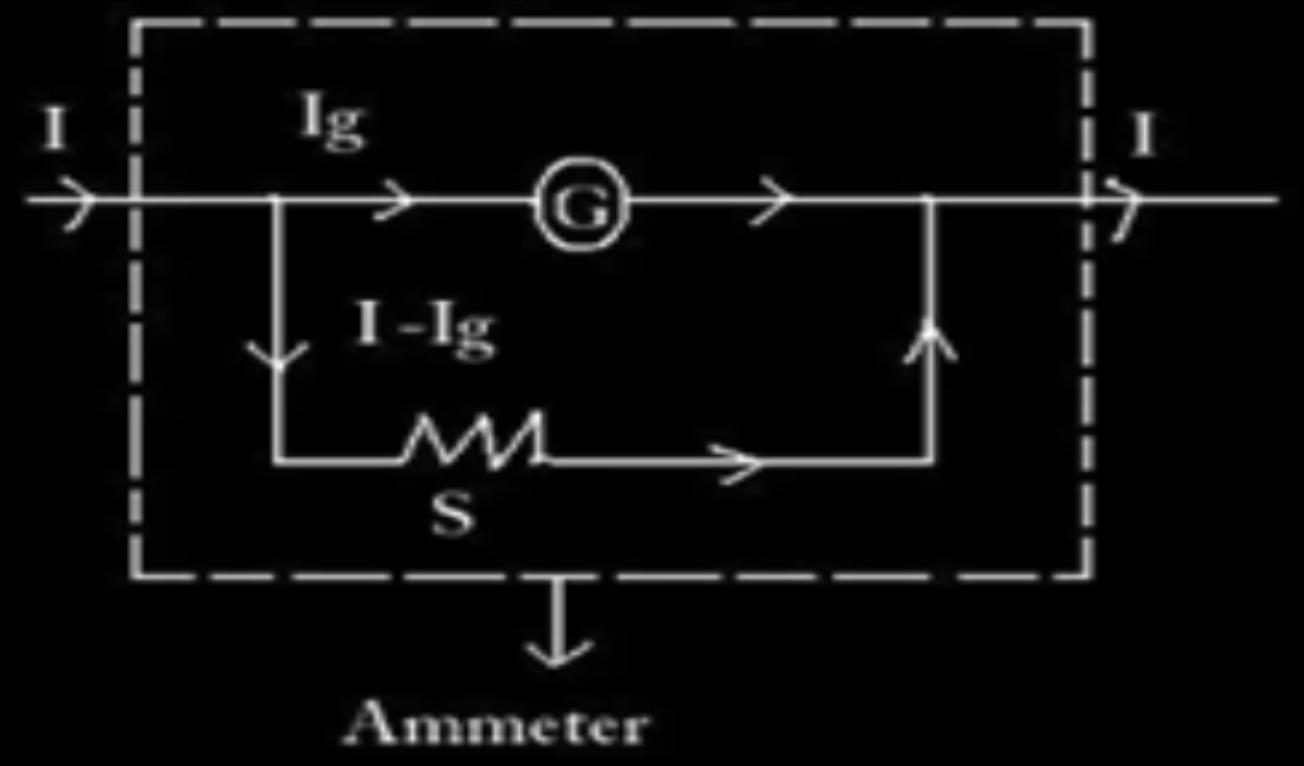
OR

Explain how a galvanometer can be converted into an ammeter of a given range. Derive an expression for shunt resistance and current for full scale deflection. Find the effective resistance of the ammeter.

Principle -	1
Working & expression for deflection -	1
Current Sensitivity -	1

Conversion of galvanometer into ammeter -	1
Expression for shunt resistance -	1
Effective resistance of ammeter -	1

Principle - A current carrying coil experiences a torque in a magnetic field.	1
Working - When current is passed through the coil torque produced is $\tau = NIAB \sin\theta = NIAB \quad (\quad \theta = 90^\circ)$	½
Restoring torque $\tau' = k\phi$ At equilibrium $\tau = \tau'$ $NIAB = k\phi$	½
$\phi = \left(\frac{NAB}{k}\right) I$	
$Current\ sensitivity: \frac{\phi}{I} = \frac{NAB}{k}$	1
[Alternatively: It is deflection per unit current]	

A galvanometer is converted into an ammeter when a suitable shunt resistance is connected in parallel with the galvanometer.	½
	½
[Award full one mark even if no figure drawn or only the figure is drawn]	

$G I_g = (I - I_g)S$	½
$S = (\frac{I_g}{I - I_g})G$	½
$R_{eff} = \frac{GS}{G+S}$	1

(a) Briefly explain how a galvanometer is converted into an ammeter.

(b) A galvanometer coil has a resistance of 15 Ω and it shows full scale deflection for a current of 4 mA. Convert it into an ammeter of range 0 to 6 A.

i) Conversion of a galvanometer into an ammeter	1½
ii) Formula for shunt	½
iii) Calculation and Result	1

a)	By connecting a small resistance called shunt (S) in parallel to coil of the galvanometer The value of S is related to the maximum current (I) to be measured as $S = I_g G / I - I_g$. (Note : If the student just draws the diagram, full marks may be awarded).	1½
b)	$G = 15\,\Omega$ $I_g = 4 \times 10^{-3} A$ $I = 6A$	
	$I_g G = (I - I_g) S$	½
	$S = \dfrac{I_g G}{I - I_g} = \dfrac{4 \times 10^{-3} \times 15}{6 - 4 \times 10^{-3}}$	½
	$= 0.01\,\Omega$	½

(a) Briefly explain how a galvanometer is converted into a voltmeter.

(b) A voltmeter of a certain range is constructed by connecting a resistance of 980 Ω in series with a galvanometer. When the resistance of 470 Ω is connected in series, the range gets halved. Find the resistance of the galvanometer.

i)	Conversion of a galvanometer into a voltmeter	1½
ii)	Formula	½
iii)	Calculation and results	1

a)	A galvanometer may be converted into voltmeter by connecting a high value resistance R in series with coil of the galvanometer. The value of (R) is related to the maximum voltage (V) to be measured as $R = \frac{V}{I_g} - G$ Note - Award full marks if a student just draws the labelled diagram.	1½
b)	$I_g = \frac{V}{R_g + R}$	½
	$\frac{V}{R_g + 980} = \frac{V}{2(R_g + 470)}$	½
	$2R_g + 940 = R_g + 980$	
	$R_g = 40\Omega$	½

How is a galvanometer converted into a voltmeter and an ammeter ? Draw the relevant diagrams and find the resistance of the arrangement in each case. Take resistance of galvanometer as G.

How galvanometer is converted in to a voltmeter and an Ammeter	½ + ½
Diagram for conversion of galvanometer into a voltmeter and an Ammeter.	½ + ½
Resistance of each arrangement	½ + ½

A galvanometer is converted into a voltmeter by connecting a high resistance 'R' in series with it.	½
A galvanometer is converted into an ammeter by connecting a small resistance (called shunt) in parallel with it.	½
Resistance of voltmeter, $R_V = G + R$	½
Resistance for Ammeter, $R_A = \frac{G\, r_s}{G + r_s}$	½